What does it mean to do work?

? A porter carries a suitcase along a level platform all day and is exhausted. How much work did he do on the suitcase?

1

One product, no direction left

Work is a scalar built from two vectors: the force and the displacement. Only the part of the force along the displacement moves energy into or out of the body; the perpendicular part changes direction and earns nothing.

$$W = \mathbf{F} \cdot \mathbf{d} = Fd\cos\theta$$

→ the sign is not a direction: it says whether energy enters or leaves

2

The three signs

Positive work: the force has a component along the motion, so the body speeds up. Negative work: the force opposes the motion, and the body slows. Zero work: the force is perpendicular to the displacement, whatever its size.

$$\theta < 90^\circ : W > 0 \qquad \theta = 90^\circ : W = 0 \qquad \theta > 90^\circ : W < 0$$

3

Work is not effort

The first trap of the chapter: biology and physics use the word differently. A wall pushed for an hour receives zero work; a satellite in a circular orbit is pulled by gravity for ever and gravity does zero work on it each orbit.

→ if there is no displacement along the force, the work is zero however hard it feels

4

A worked pair of pulls

A 20 kg crate is dragged 6 m by a 120 N rope at 35° above the horizontal, with friction 50 N. Find the work of the rope, of friction, and the net work.

$$W_F = 120(6)\cos 35^\circ = 590 \text{ J}$$

$$W_f = -50(6) = -300 \text{ J}, \qquad W_{\text{net}} = 290 \text{ J}$$

→ gravity and the normal do nothing: both are vertical, the displacement is horizontal

KEY RESULT

the definition

$$W = Fd\cos\theta$$

F and d in the same frame, theta measured between them

TRAP

selling effort as work

Holding, pushing a wall, carrying level: zero work. Muscles do internal work, the object does not gain energy.

NUMBERS

a check

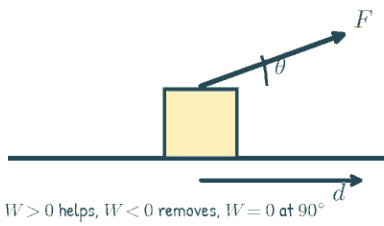
F = 50 N at 60° over 4 m

W = 50 x 4 x 0.5 = 100 J

work_signs

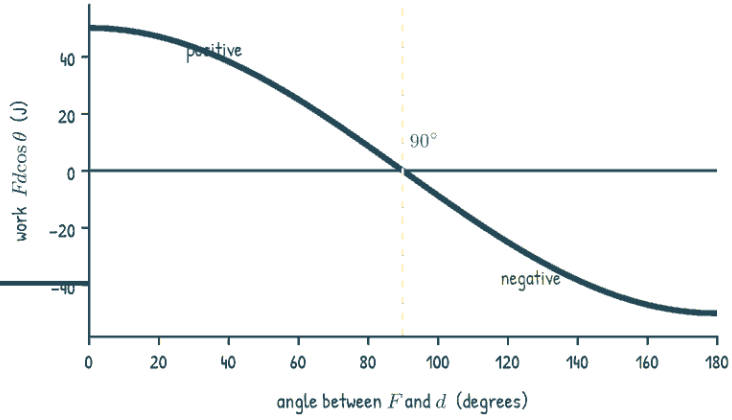
$W = Fd\cos\theta$

only the component along d counts



$W > 0$ helps, $W < 0$ removes, $W = 0$ at 90°

one curve, two signs



KEEP / work is the component of force along the displacement, with a sign

When the force changes, what replaces F d?

? A spring stiffens as you pull it, and a gas pushes less hard as it expands. What is the work when the force is a curve?

1 **Slice the path**

Over a short step the force is nearly constant, so the work is F dx. Adding the slices gives an integral, and the integral of a plotted curve is an area.

$$W = \int_{x_i}^{x_f} F(x)\,dx = \text{area under the } F\text{–}x \text{ graph}$$

2 **Areas below the axis subtract**

A force that reverses sign gives back energy. Pushed out and pulled back, the two areas cancel: the net work is the algebraic sum, not the shaded total.

$$W_{\text{net}} = \int F\,dx \quad \text{counted with sign, not } \int |F|\,dx$$

3 **The spring sets the template**

For F = kx the graph is a straight line from the origin, so the area is a triangle: half the final force times the extension. Every elastic-energy result is this triangle.

$$W = \int_0^{x} kx\,dx = \tfrac{1}{2}kx^2$$

Read it as an average: the force runs from 0 to kx, so its average is kx/2.

4 **A worked area**

A force F = 6x N acts from x = 0 to x = 2 m, then falls linearly to zero at x = 4 m. Find the total work.

$$W_1 = \int_0^{2} 6x\,dx = 12\ \text{J}, \qquad W_2 = \tfrac{1}{2}(12)(2) = 12\ \text{J}$$

$$W = 24\ \text{J} \ (\text{the two areas add, } F \text{ never reversed})$$

→ the second part is a triangle: 12 N at the start of it, zero at the end

KEY RESULT

variable force

$$W = \int F\,dx$$

equivalently the signed area of the F-x graph

GATE

check the shape first

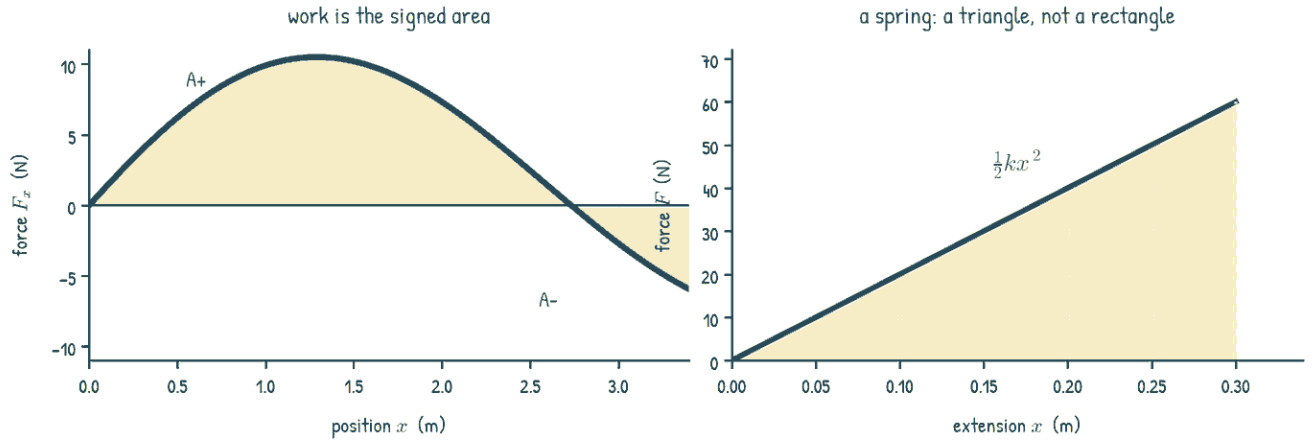
Rectangle, triangle, or a curve you can integrate? Sketch it before writing any integral.

TRAP

using the peak force

For a linear spring the work uses the average force kx/2, not the final force kx. Using kx doubles every spring answer.

fx_area



KEEP / integrate the force along the path: the answer is a signed area

Which forces actually change the energy?

? A block slides down a rough incline. Four forces act. Which of them can put energy in, which can only take it out, and which is powerless?

1

The four standard answers

Gravity, normal, friction and the spring cover nearly every JEE question. Each has a fixed signature: gravity stores and returns, the normal on a fixed surface does nothing, friction only removes, and a spring does both.

$$W_g = mg(y_i - y_f), \quad W_N = 0, \quad W_f = -f_k d, \quad W_s = -\tfrac{1}{2}kx^2$$

2

Why the normal is usually zero

The normal force is perpendicular to the surface, and on a fixed surface the body slides along it, so the displacement is perpendicular to the normal at every instant. Zero dot product, zero work.

The moment the surface moves, the argument dies. On a wedge that slides, the block displacement is not along the face, the normal is no longer perpendicular to it, and the normal does work. That case is OLY 05.

→ the normal is a zero only while the surface is at rest

3

Friction is the only one-way street

Kinetic friction opposes the relative sliding, so it always does negative work on each sliding body, and the sum on the pair is negative. That negative work is the heat.

$$W_f^{\text{pair}} = -f_k \, s_{\text{rel}}$$

→ push a block round a closed loop and friction still removes energy; gravity returns all of it

4

A worked audit

A 5 kg block slides 4 m down a 30° incline with mu_k = 0.25. Find the work of each force and the final speed from rest.

$$W_g = 5(9.8)(4)(0.5) = 98 \text{ J}, \qquad W_f = -0.25(42.4)(4) = -42.4 \text{ J}$$

$$W_{\text{net}} = 55.6 \text{ J} \;\Rightarrow\; v = \sqrt{2(55.6)/5} = 4.72 \text{ m/s}$$

→ the normal does zero work on a fixed incline, however large it is

KEY RESULT

the ledger of standard forces

gravity: path-free. normal: zero on a fixed surface. friction: negative. spring: returns what it stores.

NUMBERS

2 kg, 3 m, 30°, mu 0.2

W_g = +29.4 J, W_N = 0, W_f = −10.2 J

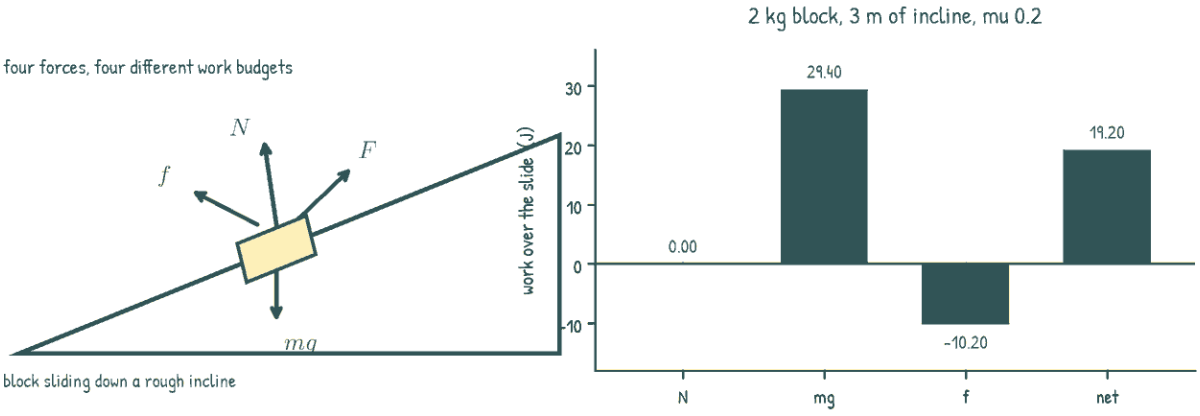
net 19.2 J = change in K

TRAP

counting a force twice

If you use the work-energy theorem, do not also add gravity as a potential. Choose: forces and W_net, or energies and conservation.

force_audit



KEEP / audit each force separately; the normal force is almost always a zero

Why is Newton enough to get the work-energy theorem?

? The theorem is usually quoted as a separate principle. Can you derive it from F = ma in one line?

1 **Convert time into space**

Newton says F = ma with a = dv/dt. Multiply both sides by the displacement dx, and use the chain rule to swap the time derivative for a speed derivative.

$$F\,dx = m\frac{dv}{dt}dx = m\,v\,dv$$

2 **Integrate both sides**

The left side accumulates work; the right side accumulates the change in half m v squared. The theorem is therefore not an extra law but a restatement of Newton along a path.

$$W_{\text{net}} = \int_{v_0}^{v} mv\,dv = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mv_0^2 = \Delta K$$

3 **What the theorem does and does not say**

It relates net work to the change in speed and is a scalar statement: no components, no angles, one number per side. It says nothing about time or direction, which is why the force method still exists.

For a system the net work includes internal work: a spring between two blocks works on both, so internal forces can change the total kinetic energy.

→ the theorem is exact in Newtonian mechanics, for any force and any path

4 **A worked theorem**

A 1200 kg car at 20 m/s brakes to rest in 55 m. Find the average braking force, then the work done.

$$W_{\text{net}} = 0 - \tfrac{1}{2}(1200)(400) = -240\ \text{kJ}$$

$$\bar{F} = \frac{240000}{55} = 4.36\ \text{kN backward}$$

→ no time was needed; the theorem is the shortest route to a force when distances are given

KEY RESULT

the theorem

$$W_{\text{net}} = \Delta K$$

one particle; for a system add the internal work

GATE

when to reach for it

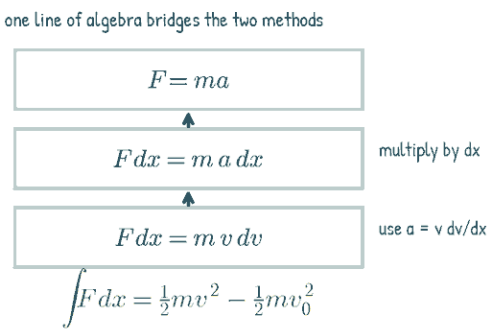
Ask for a speed or a distance and the theorem is usually shorter than F = ma. Ask for a time and it cannot help.

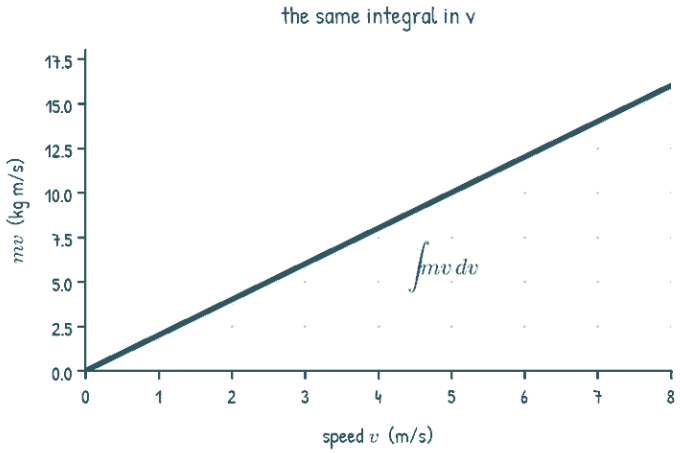
TRAP

W without the net

The theorem needs the work of every force. Work of one force equals the change in K only when it is the only one acting.

we_derivation





KEEP / multiply Newton by a displacement: the theorem is Newton integrated once over space

Whose kinetic energy is it?

? A 1 kg block moves at 4 m/s in the ground frame and 3 m/s in a train. Two observers, two kinetic energies. Is the law broken?

1

Half m v squared is not frame-free

Speed depends on the observer, so kinetic energy does too. A block at rest in a train has 8 J in one frame and 24.5 J in another, and neither observer is wrong.

$$K_{\text{new}} = K + \mathbf{p}\cdot\mathbf{V} + \tfrac{1}{2}mV^2$$

→ the cross term carries the difference between the two frames

2

The theorem survives the change

In each frame the net work computed in that frame equals the change in kinetic energy in that frame. Nothing is violated, but the numbers differ because the forces act through different displacements.

This is why a statement like "the collision released 50 J" needs a frame. Use the centre-of-mass frame and the internal energy is unambiguous; that is the whole point of the CM decomposition in OLY 01.

3

The rule for exams

Fix the frame at the start, write it at the top of the page, and keep the ground frame unless the question names another. Mixing two frames in one energy equation is the single commonest way to lose a JEE Advanced question.

→ energy differences within one frame are always safe; energies across frames are not

4

A worked frame shift

A 2 kg ball moves at 5 m/s in the ground frame; a train moves at 3 m/s the same way. Compare the kinetic energies and the work to stop it in each frame.

$$K_{\text{ground}} = 25\ \text{J}, \qquad K_{\text{train}} = \tfrac{1}{2}(2)(2)^2 = 4\ \text{J}$$

$$W_{\text{stop}} = -25\ \text{J} \text{ vs } -4\ \text{J} \text{ depending on the frame}$$

→ both are correct: the stopping distance differs between the frames too

DEFINE

kinetic energy

$$K = \tfrac{1}{2}mv^2$$

a scalar, but relative to a chosen frame

TRAP

mixing frames mid-solution

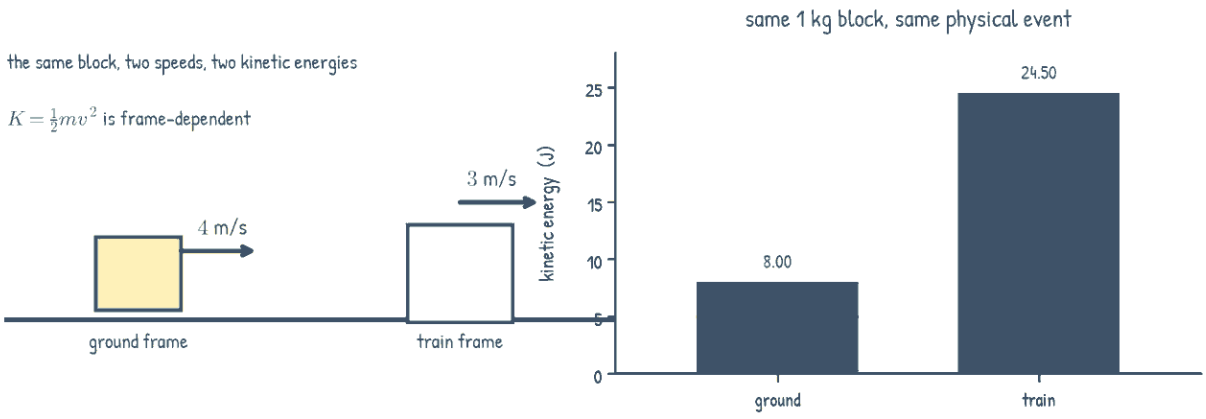
Add a train-frame energy to a ground-frame work and the answer is meaningless. Name the frame first.

INSIGHT

why the CM frame is special

In the CM frame the total momentum is zero, so the kinetic energy is purely internal and can all be converted.

ke_frames



KEEP / kinetic energy is frame-dependent, so a work statement must name its frame

Where does the energy go when friction stops a block?

? A block slides to rest. The work-energy theorem says the kinetic energy vanished. Energy is conserved, so where is it?

1

The ledger with a heat line

Mechanical energy is not conserved when friction acts, but total energy is. Write the heat as an explicit line in the budget instead of pretending it disappeared.

$$K_i + U_i = K_f + U_f + Q, \qquad Q = f_k\, d$$

2

Why d is the sliding distance

The heat is the friction force times the distance the surfaces rub against each other. On a block sliding over a fixed floor that is just the block displacement. On a belt it is not: the belt moves too, and the heat is f_k times the relative displacement.

$$Q = f_k\, s_{\text{rel}} \quad \text{not} \quad f_k \times (\text{either body displacement})$$

→ on a belt the extra heat is exactly the work done by the motor that the block did not get

3

A block that returns

Send a block up a rough incline and let it come back. Gravity returns all its work over the round trip; friction removes energy on both legs, so the return speed is lower. The difference between the two is twice the friction work.

$$v_{\text{back}}^2 = v_0^2 - 4\mu_k g \cos\theta\, s$$

4

A worked heat budget

A 3 kg block at 8 m/s slides on mu_k = 0.35 until it stops. Find the distance, the heat, and the temperature rise of a 0.4 kg steel slider taking half the heat (c = 450 J/kg K).

$$d = \frac{64}{2(0.35)(9.8)} = 9.3\ \text{m}, \qquad Q = 96\ \text{J}$$

$$\Delta T = \frac{48}{0.4(450)} = 0.27\ \text{K}$$

→ all 96 J is still there, just no longer mechanical

KEY RESULT

the heat line

$$Q = f_k\, d, \qquad K_i + U_i = K_f + U_f + Q$$

d is the relative sliding distance

NUMBERS

a check

2 kg at 6 m/s, mu_k 0.3

d = 36/(2 x 2.94) = 6.1 m, Q = 36 J

TRAP

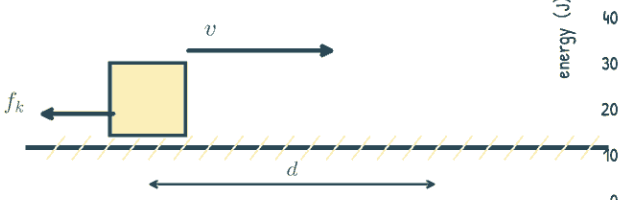
calling the energy lost

It is not lost: it is thermal energy in the two surfaces. Say so, and the bookkeeping stays honest.

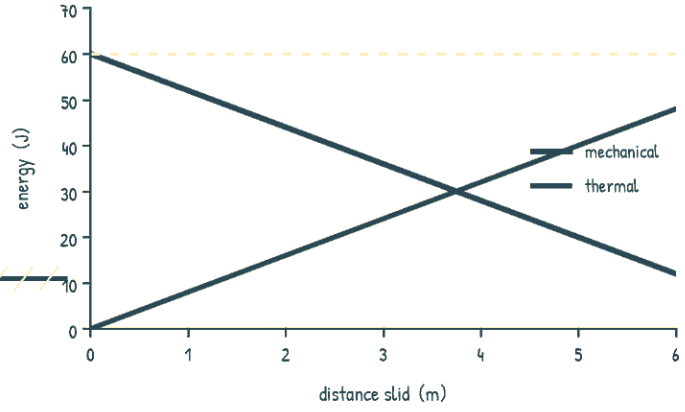
friction_ledger

friction does negative work on the block

$W_f = -f_k d$, and the same $f_k d$ appears as heat



the total is flat, the split is not



KEEP / friction turns mechanical energy into heat at f_k times the sliding distance

What is power, and why is it a dot product?

? Two motors lift the same load through the same height; one takes twice as long. The work is identical. What differs?

1 ## Rate, not amount

Power answers how fast energy moves. Average power is work over time; instantaneous power is the derivative, which turns into a dot product once the displacement is written as velocity times time.

$$\bar{P} = \frac{W}{\Delta t}, \qquad P = \frac{dW}{dt} = \mathbf{F}\cdot\mathbf{v} = Fv\cos\theta$$

2 ## The engine only feels the component along the motion

Push a cart at an angle and only F cos(theta) drives it, so only that part costs power. Push perpendicular to the motion and the power is zero however hard you push, which is the same statement as zero work.

→ P = F v is the single most used formula in vehicle and pump problems

3 ## Units and the horsepower

The watt is one joule per second. The commercial horsepower is 746 W, and the metric horsepower is 735 W; JEE questions mean 746 unless they say otherwise. A kilowatt-hour is 3.6 MJ, which is energy, not power.

$$1\ \text{hp} = 746\ \text{W}, \qquad 1\ \text{kWh} = 3.6\times10^{6}\ \text{J}$$

4 ## A worked power pair

A crane lifts 800 kg through 15 m in 20 s. Find the average power, and the instantaneous power at a steady 0.75 m/s.

$$\bar{P} = \frac{800(9.8)(15)}{20} = 5.88\ \text{kW}$$

$$P = Fv = 7840(0.75) = 5.88\ \text{kW}$$

→ they agree because the speed is constant: average equals instantaneous for uniform motion

KEY RESULT

power

$$P = \mathbf{F}\cdot\mathbf{v} = Fv\cos\theta$$

average: W over the interval

GATE

what the question gives

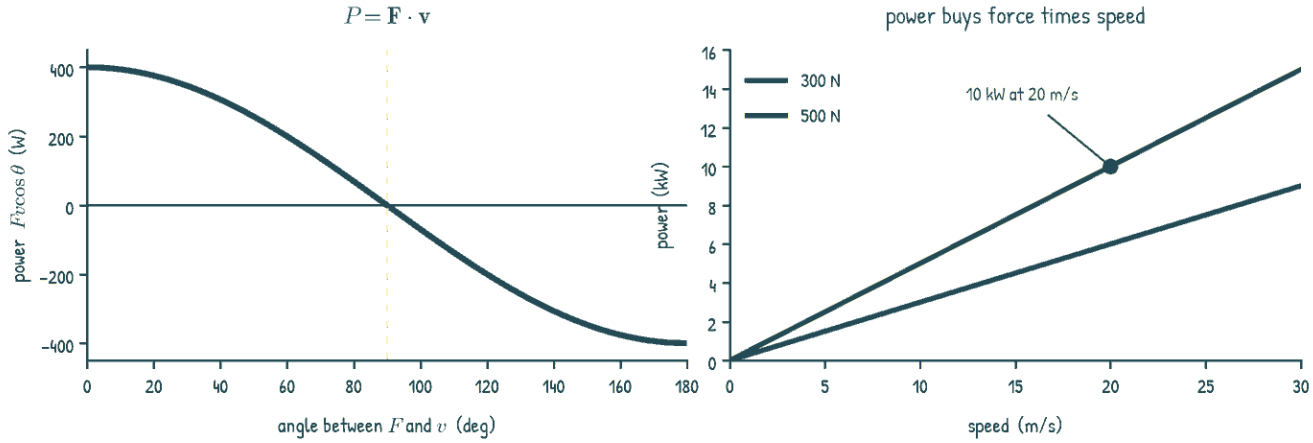
Work and time: average power. Force and speed: instantaneous power. Neither: you need one of them first.

TRAP

kWh is energy

A 100 W bulb for 5 hours uses 0.5 kWh = 1.8 MJ. The kilowatt-hour is not a unit of power.

power_curve



KEEP / power is work per unit time, and at a given speed it is force times velocity

What happens when the power is held constant?

? A car with a 100 kW engine at full throttle does not accelerate at a constant rate. What does its speed do?

1

Power fixed means force falls

At constant power the driving force is P/v, so as the car speeds up the force shrinks. Newton then gives an acceleration that falls like 1/v, and the speed grows without bound but ever more slowly.

$$F = \frac{P}{v} \quad \Rightarrow \quad a = \frac{P}{mv}$$

2

Integrate for the speed

Write a = v dv/dx or use energy directly: the work done in time t is Pt, so the kinetic energy is Pt. That gives the speed as the square root of time, and the time to reach a speed as the square of that speed.

$$v = \sqrt{\frac{2Pt}{m}}, \qquad t = \frac{mv^2}{2P}$$

→ a constant-power start is not a constant-acceleration start

3

Where it stops being ideal

Real engines reach a torque limit at low speed, so the launch is closer to constant force, and a speed limit at high speed, so the top end is closer to constant speed. The constant-power curve is the middle regime, and it is the one JEE questions use.

4

A worked spin-up

A 1500 kg car with a 60 kW engine starts from rest at constant power. Find the speed at 5 s, 20 s, and the time to 30 m/s.

$$v(5) = \sqrt{2(60000)(5)/1500} = 20 \text{ m/s}$$

$$v(20) = 40 \text{ m/s}, \qquad t(30) = \frac{1500(900)}{2(60000)} = 11.3 \text{ s}$$

→ t grows as v squared, so the last 10 m/s needs more time than the first 20

KEY RESULT

constant power

$$v = \sqrt{2Pt/m}, \qquad t = mv^2/(2P)$$

force P/v, acceleration P/(mv)

NUMBERS

a check

800 W on 1000 kg

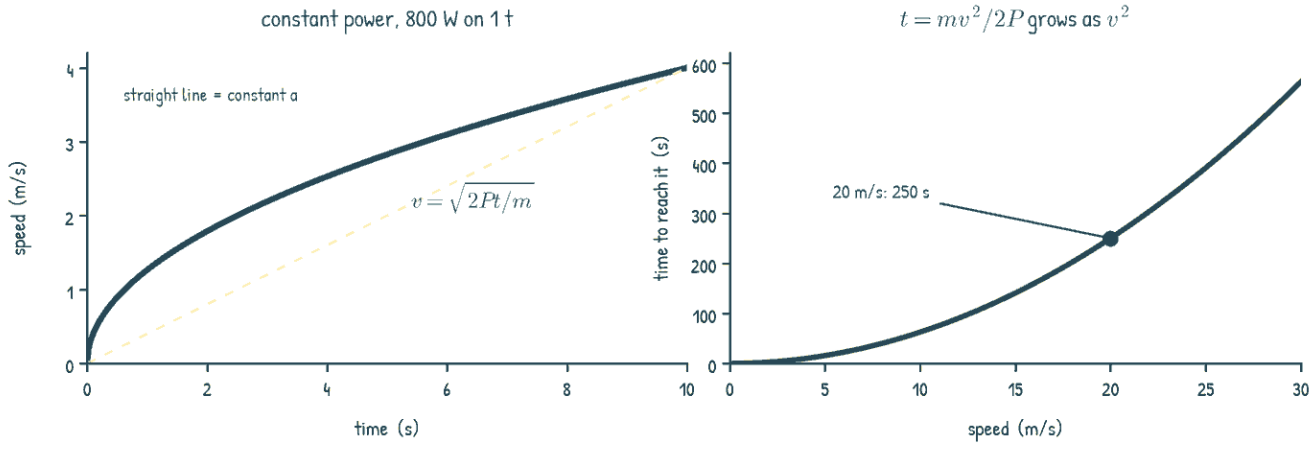
v(10 s) = 4.0 m/s; 20 m/s needs 250 s

TRAP

applying v = at

Constant acceleration is the wrong model here. Use energy, or integrate a = P/(mv).

constant_power



KEEP / at constant power the acceleration must fall as the speed rises

Why does a hill cost so much more than a straight road?

? A car cruising at 20 m/s needs 10 kW on the flat. On a 5° climb it needs much more. Where does the extra power go?

1

Separate the two jobs

On the flat the engine only fights resistance. On a slope it must also lift the car, and lifting costs power at a rate equal to the weight component along the slope times the speed.

$$P = (f + mg\sin\theta)\,v$$

2

Numbers make the point

For a 1000 kg car with 500 N of drag, the flat requirement at 20 m/s is 10 kW. A 5° slope adds 1000 x 9.8 x 0.087 = 855 N, so the power jumps to 27 kW, nearly three times as much for the same speed.

$$P_{\text{flat}} = 10.0\ \text{kW}, \quad P_{5^\circ} = 27.1\ \text{kW}$$

→ most of the extra power goes into height, not into speed

3

The same algebra for pumps

A pump lifting water at a volume rate Q to a height h needs rho Q g h of useful power, and more at the shaft because of losses. Divide the useful power by the efficiency to get the input the motor must supply.

$$P_{\text{useful}} = \rho Q g h, \qquad P_{\text{in}} = \frac{\rho Q g h}{\eta}$$

4

A worked climb

A 1200 kg car climbs a 6° slope at 25 m/s against 600 N of drag. Find the power at the wheels and the extra power the slope costs.

$$P = (600 + 1200(9.8)\sin 6^\circ)(25)$$

$$= (600 + 1230)(25) = 45.8\ \text{kW}$$

→ the slope alone is 30.8 kW of it, two thirds of the total

KEY RESULT

climbing power

$$P = (f + mg\sin\theta)v$$

the sine, not the tangent: work is against the weight component

NUMBERS

a check

1000 kg, 5°, 20 m/s, f = 500 N

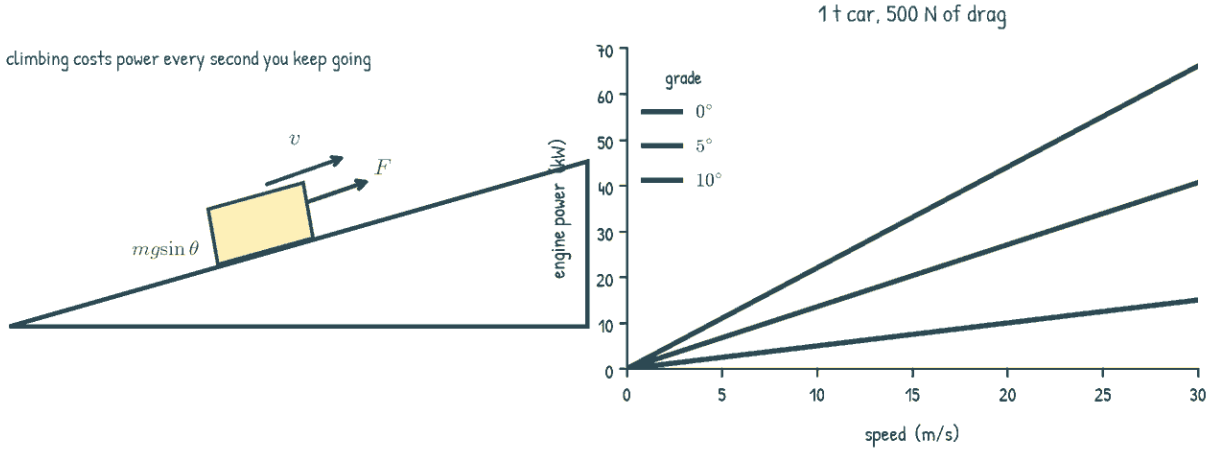
P = (500 + 855) x 20 = 27.1 kW

TRAP

using mg tan theta

The force along a slope is mg sin theta. The tangent appears in banked-road problems, not in climbing power.

slope_power



KEEP / climbing at speed v costs mgv sin theta just to gain height

What makes a force conservative?

? Gravity and the spring let you keep a potential. Friction does not. What single test separates them?

1 ## Two equivalent tests

A force is conservative if its work round every closed loop is zero, equivalently if the work between two points is path-free. The tests imply each other: go out one way and back another and the loop integral is the difference.

$$\oint \mathbf{F}\cdot d\mathbf{r} = 0 \quad \Longleftrightarrow \quad W_{A\to B} \text{ path} - \text{free}$$

2 ## Which forces pass

Gravity, the spring and every central force f(r) r-hat pass. Friction, drag and any speed-dependent force fail, because reversing the motion keeps the force against the travel.

The failure is not a technicality: a brake cannot store energy at all.

→ a force that depends on speed can never be conservative

3 ## Why we care

Only a conservative force has a potential, because only then is the work a difference of a function of position. That is what makes the energy method possible, and why friction is always a separate loss line.

$$W_{\text{cons}} = -\Delta U \quad \Longleftrightarrow \quad \oint \mathbf{F}\cdot d\mathbf{r} = 0$$

4 ## A worked loop test

A force F = (3y, 0) N acts in the plane. Compute the work round the unit square from the origin and decide.

$$\left| \oint F\cdot dr \right| = 3(1)(1) = 3 \text{ J} \neq 0$$

$$\partial_x F_y - \partial_y F_x = 0 - 3 = -3 \neq 0 \text{ so no potential}$$

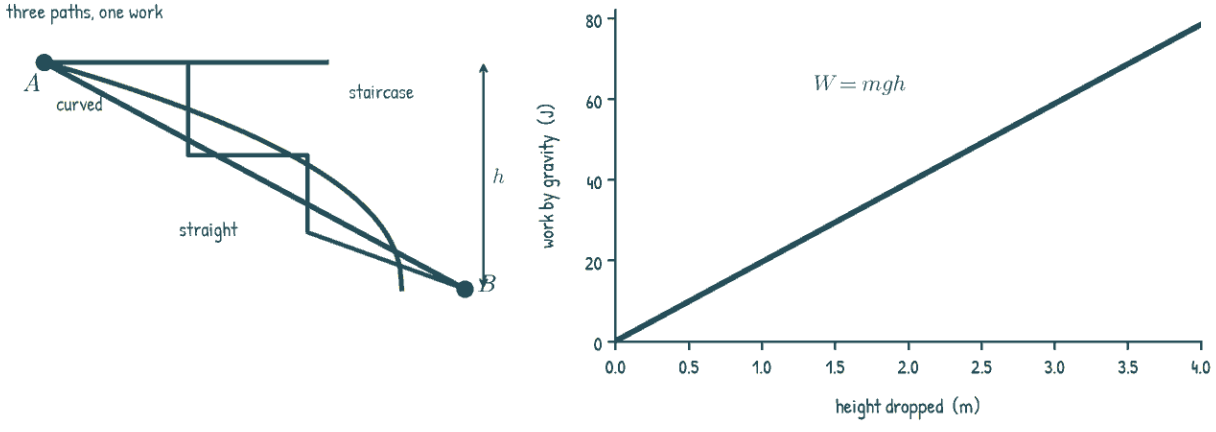
→ the two tests agree, as they must: the field is not a gradient

KEY RESULT
the test
Loop work zero, or path-independent work. Either one, and a U exists.

GATE
spot the failure fast
Does the force depend on speed or on the direction of travel? Then it is not conservative.

TRAP
gravity is not always free
Near the Earth U = mgh; far from it U = -GMm/r. Both are conservative, but the linear form is only valid while g is constant.

loop_test



KEEP / conservative means the work depends only on the endpoints, so a potential exists

Why can the zero of potential energy be anywhere?

? A book on a shelf has 20 J of potential energy, or 5 J, or −100 J, depending on where you put the zero. Which is right?

1 ## Define it by the work

Potential energy is the work a conservative force would do on the way to the reference point, with a minus sign. Because only differences enter the energy equation, the additive constant is free.

$$U(x) = -\int_{x_0}^{x} F\,dx + U(x_0)$$

2 ## Choose the zero where it simplifies

Put the zero at the lowest point for a pendulum, at the ground for a falling body, at the natural length for a spring. The physics never changes; only the arithmetic does.

$$U_g = mgh, \qquad U_s = \tfrac{1}{2}kx^2, \qquad U_g = -\frac{GMm}{r}$$

→ the minus sign in the gravitational form is a choice with U(infinity) = 0

3 ## The trap: a potential for a non-conservative force

There is no potential energy of friction. If you write one, the same point gets two different values depending on how the body arrived, and the function is not a function. In energy questions friction belongs on the loss line, never inside U.

4 ## A worked zero shift

A 2 kg book sits 1.5 m above a table, which is 0.9 m above the floor. Find U relative to the table and to the floor, and the change if it falls to the table.

$$U_{\text{table}} = 0, \qquad U_{\text{floor}} = 29.4\ \text{J}, \qquad U_{\text{floor, book}} = 47.0\ \text{J}$$

$$W_g = mg(1.5) = 29.4\ \text{J whichever zero is chosen}$$

→ only the difference is physical, so the zero is a convenience

DEFINE

potential energy

$$U = -\int F\,dx$$

defined only for conservative forces, up to a constant

NUMBERS

the spring well

k = 200 N/m, x = 0.2 m

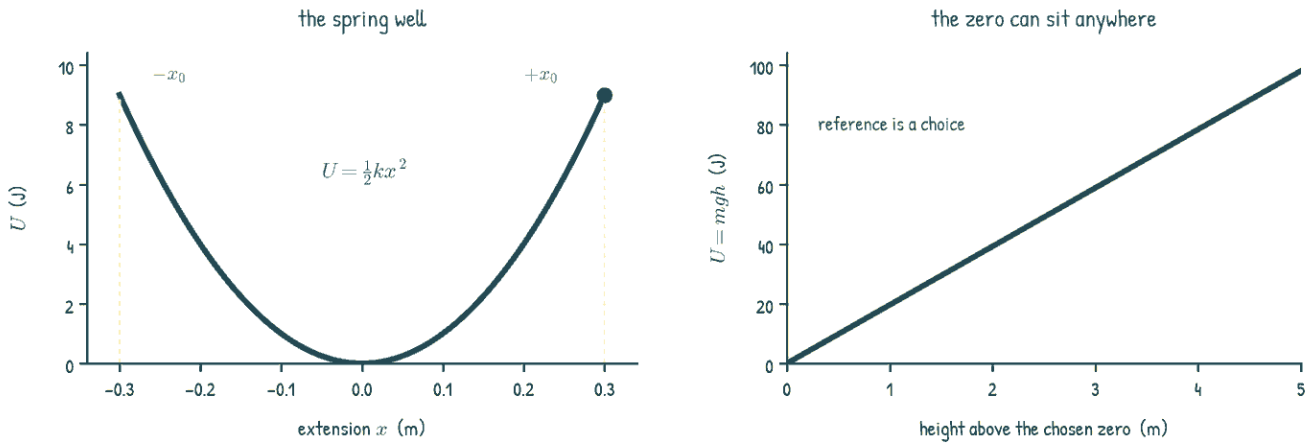
U = 0.5 x 200 x 0.04 = 4 J

TRAP

a potential for friction

Friction has no U. Write its work as −f_k d on the loss line and the bookkeeping stays truthful.

pe_definition



KEEP / only differences of potential energy are physical, so the zero is yours to choose

Why is the spring energy half k x squared?

? Pull a spring to extension x and the force has grown from zero to kx. Why is the stored energy not kx times x?

1 ## The triangle again

The force-extension graph is a straight line from the origin to (x, kx). The work done on the spring is the area of that triangle, which is half the base times the height.

$$W_{\text{on spring}} = \int_0^x kx\,dx = \tfrac{1}{2}kx^2 = U_s$$

2 ## Work by and work on

Stretching: the spring pulls back while you pull out, so the spring does negative work and stores energy. Releasing: the spring pushes in the direction of motion, does positive work, and gives the energy back. The sign flips, the magnitude does not.

$$W_{\text{spring}} = -\tfrac{1}{2}kx_f^2 + \tfrac{1}{2}kx_i^2 = -\Delta U_s$$

→ a spring is a conservative storage device, not a source

3 ## The calibration trick

Because U is quadratic, doubling the extension quadruples the energy. This is why a spring scale is linear in force but its stored energy is not, and why a small extra stretch near full compression is expensive.

→ k doubles, energy doubles; x doubles, energy quadruples

4 ## A worked spring

A spring of k = 320 N/m is compressed 12 cm and launches a 0.25 kg ball. Find the stored energy and the launch speed.

$$U = \tfrac{1}{2}(320)(0.12)^2 = 2.30\ \text{J}$$

$$v = \sqrt{2(2.30)/0.25} = 4.29\ \text{m/s}$$

→ the force at full compression is 38.4 N, but the average that did the work was 19.2 N

KEY RESULT

spring energy

$U_s = \tfrac{1}{2}kx^2$

work done by the spring is minus the change in U

NUMBERS

a check

k = 500 N/m, compressed 0.1 m

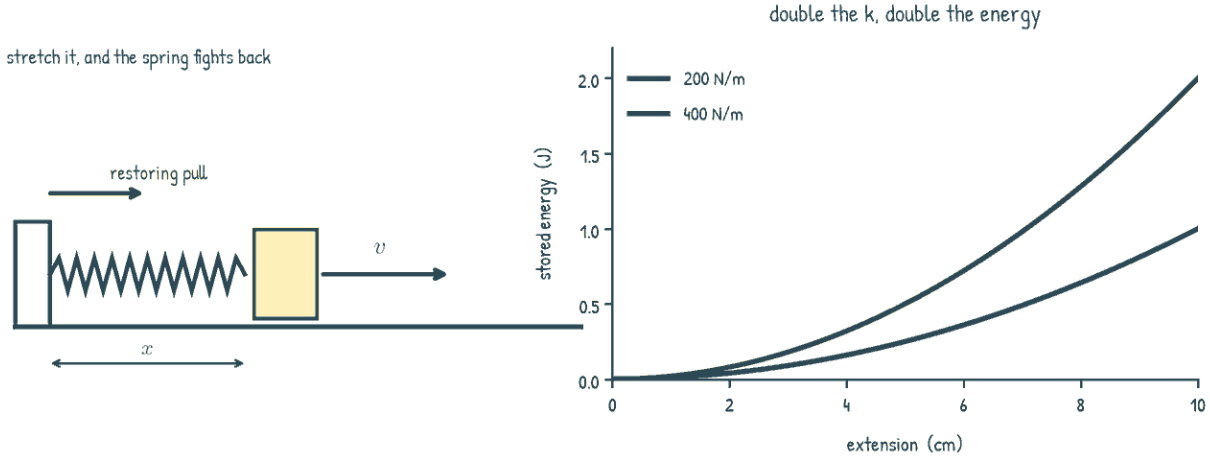
U = 0.5 x 500 x 0.01 = 2.5 J

TRAP

k x, not half k x

kx is the final force, not the average. Using it doubles the energy every time.

spring_work



KEEP / the force is not constant, so the work uses its average, which is half the final value

Does the route up a hill change the energy?

? Climb a hill by the gentle path or the steep one. The height gained is the same. Is the work by gravity the same?

1 Height is the only variable

The work by gravity is the weight times the vertical drop, whatever the path. A staircase, a ramp and a cable car between the same two points all give the same number.

$$W_g = mg(y_i - y_f) = -\Delta U_g$$

→ the horizontal part of any path is free of gravity work

2 This is why the potential exists

Path independence is exactly the condition for a potential. Gravity therefore gets a function of height, and the energy method needs no path information at all. A frictionless roller coaster is the classic beneficiary.

The same statement for a spring uses extension instead of height, and for the general gravitational field uses 1/r instead of h.

3 Where the linear form breaks

U = mgh assumes g is constant, which fails once the height is a noticeable fraction of the Earth radius. Beyond that, use −GMm/r, and the escape speed follows from setting the total energy to zero. OLY 06 does that arithmetically.

$$mgh \text{ valid for } h \ll R_E, \qquad U = -\tfrac{GMm}{r} \text{ otherwise}$$

4 A worked path comparison

A 70 kg hiker climbs 800 m of altitude along a 2.4 km trail. Compare the work against gravity with the work if the trail were 1.2 km.

$$W = mg\Delta h = 70(9.8)(800) = 549 \text{ kJ}$$

$$\text{half the trail length : the same } 549 \text{ kJ}$$

→ the trail length sets the effort, not the gravitational work

KEY RESULT

gravitational work

$$W_g = -mg\,\Delta h$$

independent of the path, so U = mgh exists

GATE

when mgh is enough

Heights under a few kilometres on Earth. Beyond that, or for orbits, switch to −GMm/r.

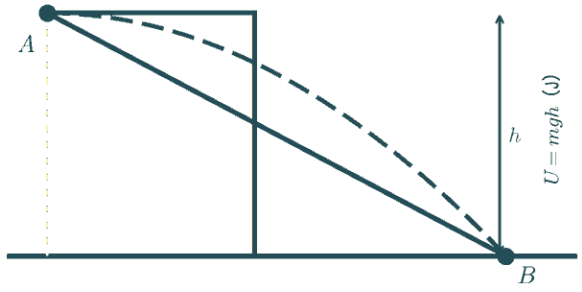
INSIGHT

the physical reading

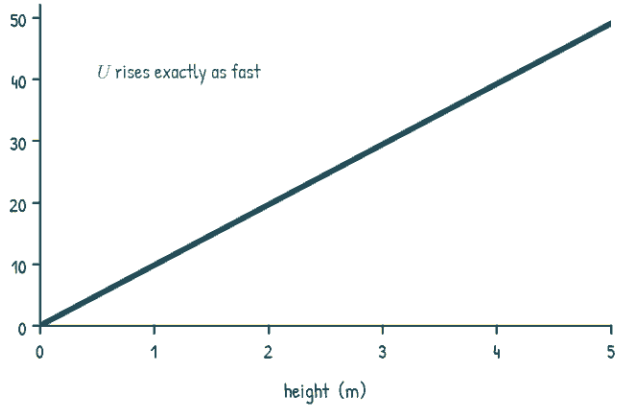
Only the vertical component of a displacement changes gravitational energy, because only that component is along the force.

gravity_paths

three paths, one work



the height is the whole story



KEEP / gravity charges by height gained, never by distance travelled

When may you write K + U = constant?

? Energy conservation is quoted as a law. Under exactly what conditions is the mechanical version true?

1 The condition, in one line

Split the net work into conservative and non-conservative parts. The conservative part is minus the change in potential, so what is left over is the change in mechanical energy.

$$W_{\text{nc}} = \Delta K + \Delta U = \Delta E$$

2 Reading the condition properly

The right-hand side being zero does not require that no friction exists; it requires that friction does no net work over that interval. A block that goes down and back up the same rough incline has friction work −2f d, so E is not constant even though the endpoints coincide.

Equally, a normal force from a moving surface can do work, and a tension at a peg that does not move cannot. Check the work, not the name of the force.

→ conservation is a statement about an interval, not about a force list

3 What to write on the sheet

Name the two states, draw the energy bars for each, and put every term that changed on the correct side. If the bars do not balance, a force was missed, and the missing term is almost always friction or a normal on a moving surface.

4 A worked conservation

A 0.6 kg ball is released from rest at the top of a 2.2 m ramp and leaves horizontally. Find its speed at the bottom and the horizontal distance after a 1.4 m drop.

$$v = \sqrt{2(9.8)(2.2)} = 6.57 \text{ m/s}$$

$$t = \sqrt{2(1.4)/9.8} = 0.535 \text{ s}, \qquad x = 3.51 \text{ m}$$

→ energy gives the speed, kinematics gives the range: the usual division of labour

KEY RESULT

conservation

$$K_i + U_i = K_f + U_f \quad \text{when } W_{\text{nc}} = 0$$

otherwise add W_nc, or subtract f_k d

TRAP

assuming conservation

A rough surface, a motor, or a moving support all break it. Look for the loss term before writing the equals sign.

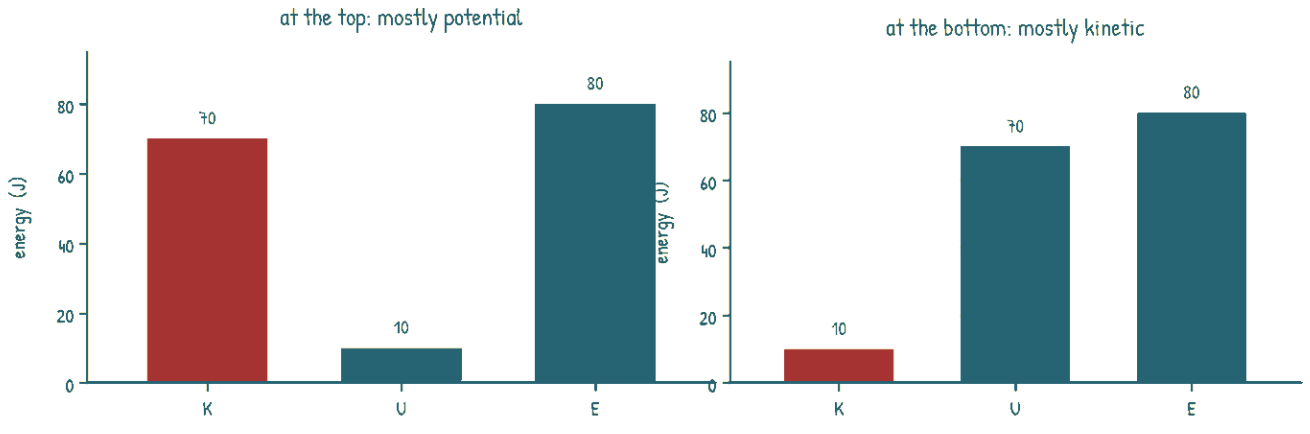
NUMBERS

a check

2 kg from 5 m on a smooth ramp

v = sqrt(2 x 9.8 x 5) = 9.9 m/s, independent of the slope

energy_ledger



KEEP / mechanical energy is constant exactly when no non-conservative force does net work

What can a potential-energy graph tell you at a glance?

? Given U(x) and one number E, how much of the motion do you already know without solving anything?

1 Draw the energy line

Put a horizontal line at the total energy. The kinetic energy is the gap between the line and the curve, so the particle can only be where the line is above U. The crossings are the turning points, where it stops and reverses.

$$K = E - U(x) \geq 0 \quad \Rightarrow \quad \text{allowed where } E \geq U$$

2 Wells, hills and stability

A minimum of U is a stable equilibrium: displaced, the particle is pushed back. A maximum is unstable. Between them the curve is a hill the particle can cross only if E reaches the top.

$$\frac{dU}{dx} = 0 \text{ and } \frac{d^2U}{dx^2} > 0 \;\Rightarrow\; \text{stable}$$

→ a well is a bound state; a hilltop is a knife edge

3 Reading speed off the graph

At any allowed x the speed is the square root of twice the gap divided by m. So the particle is fastest at the bottom of the well and slowest at the turning points, and the shape of the well is the shape of the motion.

$$v(x) = \sqrt{\frac{2(E - U(x))}{m}}$$

4 A worked well

A particle of 0.5 kg moves in U(x) = 2x² − 8x J. It is released from rest at x = 6 m. Find the turning points and the maximum speed.

$$E = U(6) = 24 \text{ J}, \qquad 2x^2 - 8x = 24 \;\Rightarrow\; x = -2,\ 6 \text{ m}$$

$$v_{\max} = \sqrt{2(24 - (-8))/0.5} = 11.3 \text{ m/s at } x = 2 \text{ m}$$

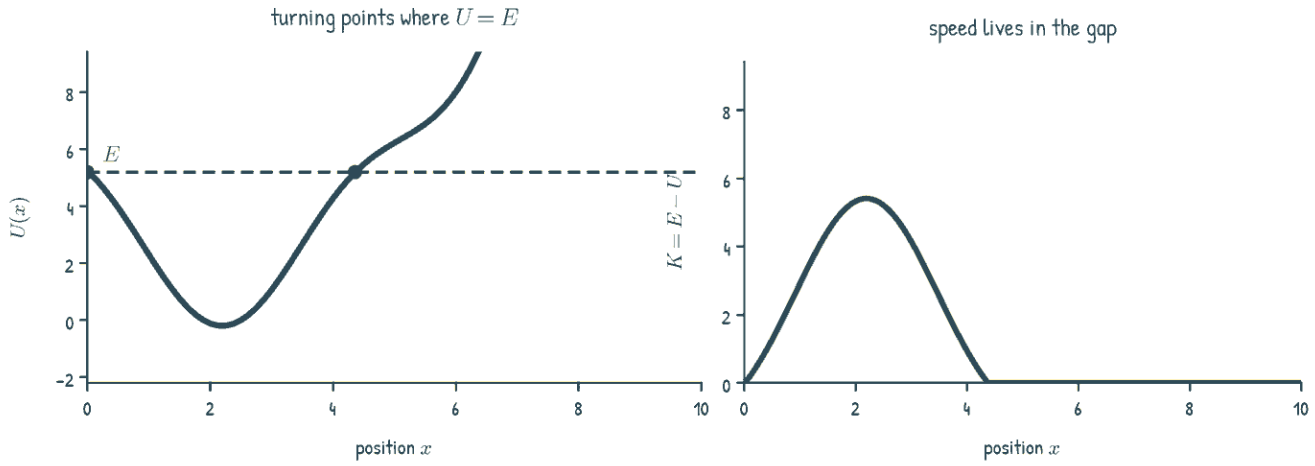
→ the well is at 2 m, the turning points are the two roots, and the motion is bounded

KEY RESULT
the three readings
crossings: turning points. minimum: stable. maximum: unstable.

NUMBERS
a check
U = 3x² − 12x + 9 J
dU/dx = 6x − 12 = 0 at x = 2 m, a minimum: stable

TRAP
calling a flat point stable
dU/dx = 0 alone is only an equilibrium. Check the second derivative, or the local shape if it vanishes.

u_landscape



KEEP / the graph is a map: where E crosses U is where the motion turns round

How do you get the force back from the energy?

? You are given U(x) as a curve. Where is the force large, which way does it point, and where is it zero?

1 **Undo the definition**

Potential energy was defined as minus the integral of the force, so differentiating brings the force back with a minus sign. The force points in the direction in which U falls.

$$F_x = -\frac{dU}{dx}, \qquad \mathbf{F} = -\nabla U$$

2 **Reading the slope**

A steep descent means a strong force; a flat region means none. At a minimum the slope is zero, so the force is zero and the particle sits in equilibrium. At the bottom of a well a small displacement produces a restoring force proportional to the displacement, which is the harmonic limit.

$$U = \tfrac{1}{2}kx^2 \;\Rightarrow\; F = -kx$$

→ the standard spring force is the gradient of the standard spring energy

3 **From a formula, not just a graph**

In three dimensions the gradient points along the steepest increase, so the force points the other way. For a central potential U(r) this gives f = -dU/dr along the radius, which is how the inverse-square law is recovered from the gravitational potential.

$$U = -\frac{GMm}{r} \;\Rightarrow\; F = -\frac{GMm}{r^2} \text{ inward}$$

4 **A worked gradient**

For U(x) = x³ − 3x J, find the force, the equilibria, and their stability.

$$F = -(3x^2 - 3) = 3 - 3x^2 \text{ N}, \qquad x = \pm 1 \text{ m}$$

$$U_{xx} = 6x: \text{ stable at } x = 1, \text{ unstable at } x = -1$$

→ at x = 1 the force pushes back toward 1 for small displacements

KEY RESULT

force from energy

$F_x = -dU/dx$

in 3-D: F = −grad U

NUMBERS

a check

U = 5x² − 10x

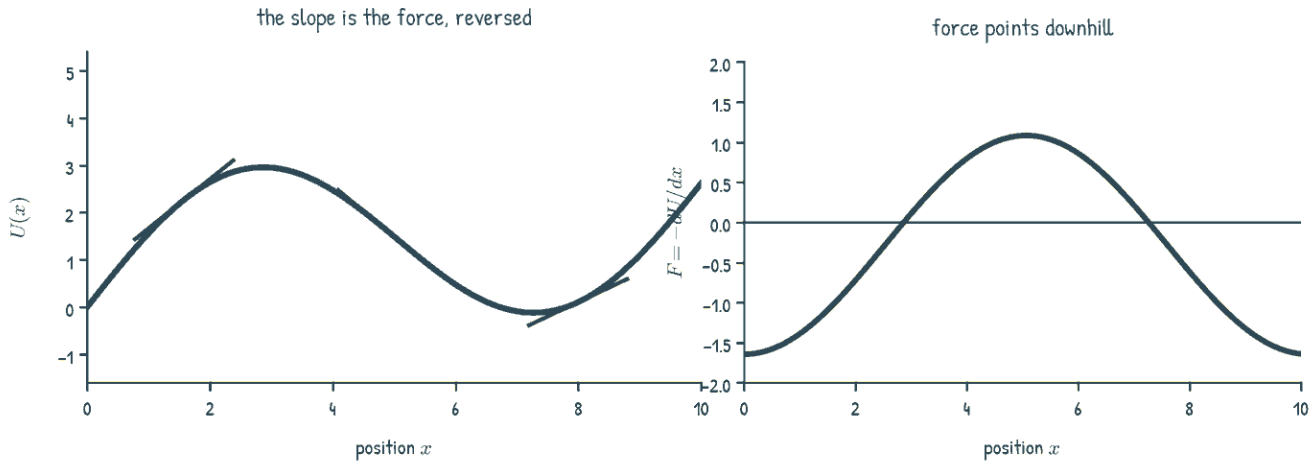
F = 10 − 10x N; zero at x = 1 m

TRAP

forgetting the minus

The force pushes the particle toward lower U. Dropping the minus reverses every direction in the answer.

force_gradient



KEEP / force is minus the slope of the potential, so it points downhill

Energy or Newton: which method does the question want?

? Two questions about the same loop: what is the speed at the top, and what is the normal force there? One is energy, one is not.

1 What each method delivers

Energy is a scalar relation between two states and gives speeds and distances without any vector work. Newton is a relation between force and acceleration at one instant and gives times, accelerations and contact forces. Neither replaces the other.

$$W_{\text{net}} = \Delta K \quad \text{vs} \quad \sum \mathbf{F} = m\mathbf{a}$$

2 The standard combination

For a bead on a loop the usual solution is two steps: energy from the release point to the point of interest for the speed, then the radial equation at that point for the normal force. Do them in that order and the algebra stays linear.

$$v^2 = v_0^2 - 2gR(1 - \cos\phi)$$

$$N = \frac{mv^2}{R} + mg\cos\phi$$

→ energy first, force second: reversing the order costs a square root in the force equation

3 Questions where energy is a dead end

Anything asking for a time, a rate, or a force at an instant. Energy gives no clock, because it says nothing about how the states are connected in time. A block on a spring has the same energy at two different extensions but reaches them at different times.

4 A worked combination

A 0.3 kg bead slides from rest at 3R above the bottom of a smooth loop of radius 0.5 m. Find the speed and the normal at the top.

$$v^2 = 2(9.8)(3-2)(0.5) = 9.8 \;\Rightarrow\; v = 3.13 \text{ m/s}$$

$$N = \frac{0.3(9.8)}{0.5} - 0.3(9.8) = 5.88 - 2.94 = 2.9 \text{ N}$$

→ energy first, radial second: both steps are one line

GATE
three-way sort
speed or distance: energy. time or acceleration: forces. force at a point: both, in that order.

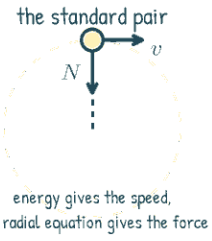
TRAP
energy for a time
It cannot give one. Use it for the speed, then integrate or use the force method for the clock.

INSIGHT
why the pair is standard
Energy removes the vector algebra; the radial equation then uses the speed it produced. Each does what it is good at.

decision_map

what the question asks decides the method

speed at B	→	energy
distance travelled	→	energy + friction
does it reach?	→	compare E with U
force at B	→	energy, then radial
time or acceleration	→	forces



KEEP / energy answers speed and distance; forces answer time, acceleration and contact force

What changes when the system has two bodies?

? Two blocks joined by a spring are pulled apart. The external work is 20 J, but the kinetic energy rises by only 12 J. Where are the other 8 J?

1 ## The theorem for a system

For a collection of particles, the net work is external plus internal, and both can change the total kinetic energy. The spring is internal, so it takes energy from the blocks and stores it.

$$W_{\text{ext}} + W_{\text{int}} = \Delta K_{\text{total}}$$

2 ## Internal conservative work is a potential

If the internal force is conservative, its work is minus the change in the internal potential. Move that term to the other side and the bookkeeping becomes external work equals the change in kinetic plus potential.

$$W_{\text{ext}} = \Delta K + \Delta U_s$$

→ this is why a two-block spring problem is one energy equation, not two

3 ## The moving-wedge warning

Internal forces also include the normal between a block and a moving wedge. That normal does work on the block and equal and opposite work on the wedge, so the pair conserves energy while neither body alone does. Energy questions about a wedge must always be asked of the pair.

→ for the pair, the normal is internal and cancels; for one body it is an external force that does work

4 ## A worked pair

Two blocks, 1 kg and 2 kg, are joined by a spring (k = 200 N/m) and pulled apart so the spring stretches 0.2 m from natural length. Released from rest, find the total kinetic energy when the stretch is 0.1 m.

$$U_i = \tfrac{1}{2}(200)(0.04) = 4.0\ \text{J}, \qquad U_f = \tfrac{1}{2}(200)(0.01) = 1.0\ \text{J}$$

$$K_{\text{total}} = 3.0\ \text{J shared between the two blocks}$$

→ the pair conserves energy; neither block does on its own

KEY RESULT

system bookkeeping

$$W_{\text{ext}} = \Delta K + \Delta U$$

with U counting every internal conservative interaction

NUMBERS

the opening question

20 J in, 12 J of K

so U_spring rose by 8 J: nothing was lost

TRAP

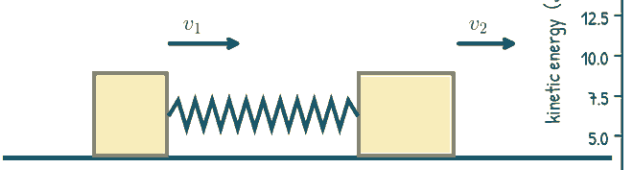
applying one-body conservation to a pair

The block on a moving wedge does not conserve its own energy. The pair does.

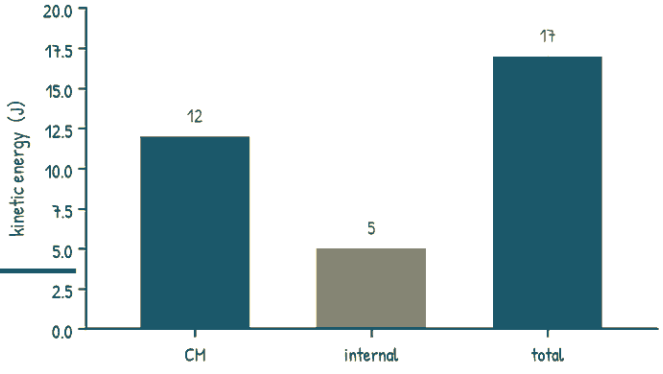
system_energy

the spring is internal, so it only stores energy

$W_{\text{ext}} = \Delta K + \Delta U_s$



the split is not a choice



KEEP / internal work does not vanish: it moves energy into the spring, not out of the system

Why does a pump need more power than the water gains?

? A pump lifts 100 kg of water per second through 20 m. The water gains 19.6 kW. The motor is rated 25 kW. Where did the rest go?

1 **The useful part**

The water gains gravitational energy at a rate equal to the mass flow times g times the height. That is the useful output, and it is fixed by the job, not by the machine.

$$P_{\text{out}} = \dot{m} g h$$

2 **The input and the ratio**

The motor supplies more because of friction in the pipes, turbulence, and the inefficiency of the pump itself. The ratio is the efficiency, and it is always below one for a real machine.

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} = \frac{\dot{m} g h}{P_{\text{in}}}$$

→ 19.6 kW out of 25 kW in is 78 per cent, which is a normal figure for a good pump

3 **The same idea elsewhere**

A car engine quotes shaft power, but only the wheel power moves the car; the difference is the drivetrain loss. A power station quotes electrical output, not heat input. Every one of these is the same ratio with different names, and in each case the missing power has a physical address.

→ efficiency questions are bookkeeping questions with one unknown

4 **A worked pump**

A pump raises 0.05 m³/s of water through 12 m. The motor draws 9 kW. Find the efficiency.

$$P_{\text{out}} = 1000(0.05)(9.8)(12) = 5.88 \text{ kW}$$

$$\eta = 5.88/9 = 0.65$$

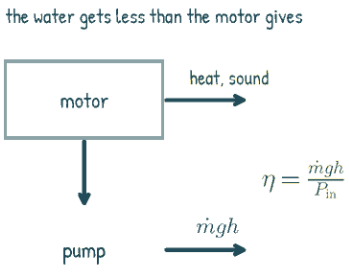
→ a third of the electrical input ends up as heat in the water and the pipes

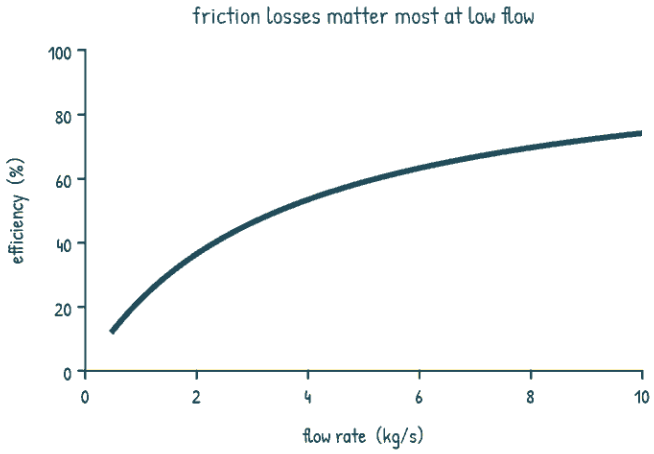
KEY RESULT
efficiency
$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}}, \qquad P_{\text{in}} = \frac{\dot{m} g h}{\eta}$$

NUMBERS
a check
100 kg/s through 20 m
P_out = 100 x 9.8 x 20 = 19.6 kW

TRAP
efficiency above one
If your answer exceeds 100 per cent, you have used the input as the output or forgotten a loss term.

efficiency





KEEP / efficiency is output over input, and every loss is a term you can name

Why does lifting a hanging chain cost more than its weight?

? A chain is pulled up from a heap at constant speed. At height x the hanging weight is lambda x g, yet the pull must exceed it. Why?

1

Force from momentum, not just weight

The hanging part needs support, and the links leaving the heap must be brought from rest to speed v every second. That is a momentum flux, and it needs a force even though the speed is constant.

$$F = \lambda x g + \lambda v^2$$

2

The energy version tells the same story

The power delivered is Fv, while the energy actually gained is the rate of increase of gravitational plus kinetic energy. The two differ by half lambda v cubed, which is the energy lost in the inelastic jerk as each link is picked up.

$$P = Fv = \lambda x g v + \lambda v^3, \qquad \frac{dE}{dt} = \lambda x g v + \tfrac{1}{2}\lambda v^3$$

→ the gap is dissipation, not an algebraic slip

3

Where students lose the mark

Using F = lambda x g alone gives a force that is too small and a work that is too small. Using energy conservation with no loss term gives an answer that is also wrong, in the opposite direction. Both errors come from ignoring the momentum flux.

→ a chain is the standard reminder that mass can be added to a moving system

4

A worked chain

A 3 m chain of 6 kg is lifted from a heap at a steady 0.8 m/s. Find the force when 2 m is up, and the power at that instant.

$$F = \lambda x g + \lambda v^2 = 2(9.8)(2) + 2(0.64) = 40.5 \text{ N}$$

$$P = Fv = 32.4 \text{ W}, \qquad \lambda v^3 = 1.0 \text{ W of it is lost}$$

→ the flux term is small at this speed but it never vanishes

KEY RESULT

the pull

$$F = \lambda x g + \lambda v^2$$

weight plus momentum flux, at constant speed

NUMBERS

a check

lambda = 0.8 kg/m, x = 2.5 m, v = 1.2 m/s

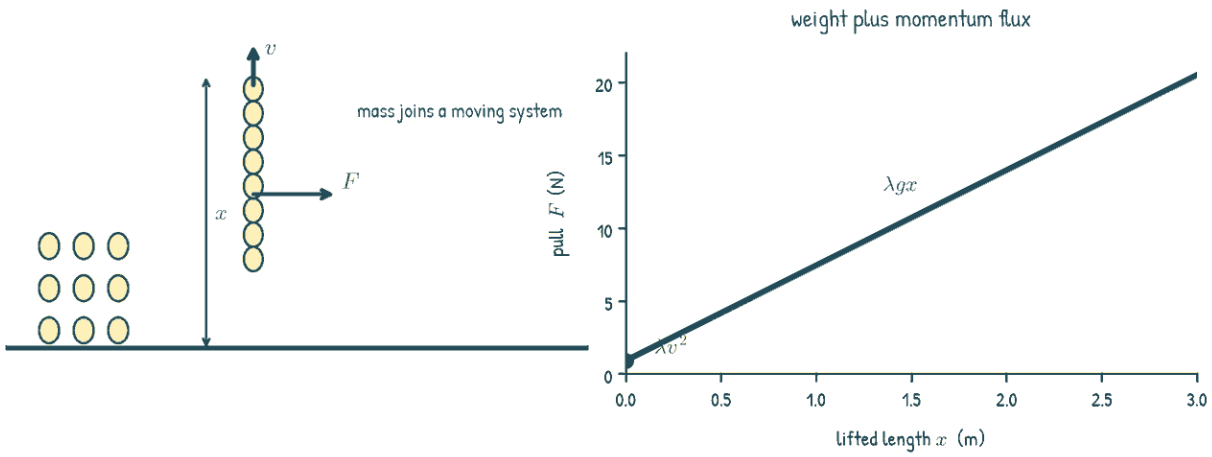
F = 19.6 + 1.15 = 20.8 N

TRAP

weight only

The flux term is small at low speed but never zero. It is the entire reason a chain feels heavier than its hanging part.

chain_energy



KEEP / new links are accelerated from rest, so the pull carries a momentum-flux term

A spring launches a block onto a rough floor. How far does it go?

? Compress a spring, release it, and let friction stop the block. Two energy stores and one loss. Which equation do you write?

1 ## One equation, three terms

At release everything is stored in the spring. At rest everything has become heat in the floor. Nothing else changes, so the spring energy and the friction work are equal.

$$\tfrac{1}{2}kx^2 = \mu_k mg\,d$$

2 ## If the block leaves the spring

The block loses contact when the spring passes its natural length, so the launch speed follows from the stored energy alone. From then on it is a decelerating block, and the stopping distance is that kinetic energy over the friction force.

$$v^2 = \frac{kx^2}{m}, \qquad d = \frac{v^2}{2\mu_k g} = \frac{kx^2}{2\mu_k mg}$$

→ the same d as the single-equation route, which is the check

3 ## The stopping condition

The block stops before the end of the rough patch only if the friction work available exceeds the stored energy. Otherwise it leaves the rough region still moving, and the answer to "how far" is "the whole patch".

$$\mu_k mg\,L \geq \tfrac{1}{2}kx^2 \;\Rightarrow\; \text{it stops inside}$$

4 ## A worked launch

A 1.2 kg block is launched by a spring of k = 400 N/m compressed 15 cm on a floor with mu_k = 0.25. Find the sliding distance after release.

$$\tfrac{1}{2}(400)(0.0225) = 4.5\ \text{J} = 0.25(1.2)(9.8)d$$

$$d = \frac{4.5}{2.94} = 1.53\ \text{m}$$

→ the block leaves the spring at its natural length; the friction acts over the whole 1.53 m

KEY RESULT

the balance

$$\tfrac{1}{2}kx^2 = \mu_k mg\,d$$

stored energy equals the friction loss

NUMBERS

a check

k = 200, x = 0.1, m = 0.5, mu = 0.1

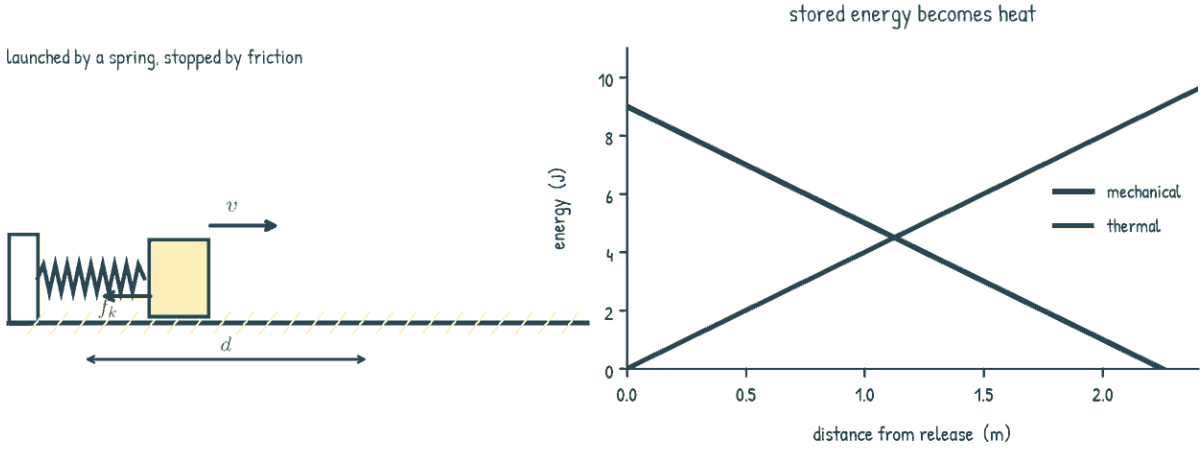
1 J = 0.49 d, so d = 2.04 m

TRAP

forgetting the block leaves the spring

The spring stops pushing at its natural length. Do not keep the spring force acting over the whole slide.

spring_friction



KEEP / stored spring energy equals the friction work, and the loss is the bridge between them

What is the whole chapter in one picture?

? Six ideas, a page of formulas and thirty-six questions. Which ideas actually carry the load?

1

Six ideas, in order

Work measures energy transfer. The theorem turns work into kinetic energy. Power is its rate. A conservative force allows a potential. Conservation is the special case with no losses. Non-conservative bookkeeping is the general case.

$$W = \int \mathbf{F} \cdot d\mathbf{r}, \quad W = \Delta K, \quad P = \frac{dW}{dt}, \quad F = -\frac{dU}{dx}$$

2

The routing table

Speed from a height: conservation. Distance on a rough surface: theorem with friction. Force at a point on a track: energy then radial equation. Power of a vehicle: F v with the resistance and the slope included. Shape of a potential: differentiate for the force, look for the wells.

→ if two routes disagree, the difference is a term you forgot to name

3

The discipline

Write the two states, list the energy terms that changed, and put every loss on the correct side. A sheet that balances is finished; a sheet that does not is missing a force, and the missing force is almost always friction or a moving support.

4

A worked routing

A 2 kg block slides 5 m down a 35° rough incline and then compresses a spring of k = 800 N/m. mu_k = 0.2. Find the compression.

$$E_{\text{avail}} = 2(9.8)(5\sin 35^\circ) - 0.2(2)(9.8\cos 35^\circ)(5)$$

$$= 56.2 - 16.1 = 40.1 \text{ J} \Rightarrow x = \sqrt{2(40.1)/800} = 0.32 \text{ m}$$

→ one energy equation with a friction line, no forces resolved twice

GATE
routing
speed or distance: energy. time or force: Newton. mixed: energy then the radial equation.

TRAP
double counting
Gravity is either a force doing work or a potential energy. Never both in one equation.

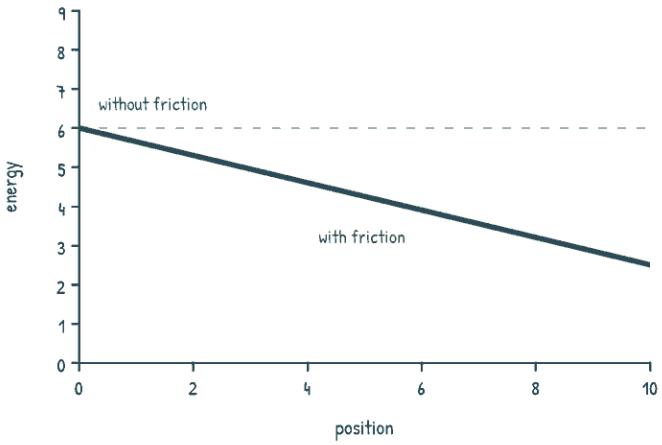
INSIGHT
the one-line summary
Every problem here is a ledger. Name the terms, sign them, and balance them.

wpe_map

six ideas, two of them bookkeeping

work F d cos	theorem $W = \Delta K$
power $F \cdot v$	potential $-\int F dx$
conservation E constant	bookkeeping $W_{\text{nc}} = \Delta E$

one line for the whole chapter



KEEP / work is the unit of account, and every result is a ledger with a named loss

Why describe rotation with an angle at all?

? Two points on a wheel, one near the rim and one near the hub, travel different distances in the same time. What do they share?

1 ## One angle, many arcs

Angular position is the angle swept by the radius from a reference direction. All points of a rigid wheel share it, so one number describes the whole body. Arc length is the radius times that angle, provided the angle is in radians.

$$\theta = \frac{s}{R}, \qquad s = R\theta$$

2 ## Rates

Differentiate: angular velocity is the rate of change of the angle, and tangential speed is radius times angular velocity. The same angular velocity gives a larger speed further out, which is why the rim of a wheel moves faster than the hub.

$$\omega = \frac{d\theta}{dt}, \qquad v = R\omega$$

→ radians are required: in degrees the factor pi/180 would have to be carried everywhere

3 ## Period, frequency and rpm

One revolution is 2 pi radians, so period and frequency follow immediately. Revolutions per minute must be converted before use: divide by 60 and multiply by 2 pi. Using rpm directly in v = R omega is the commonest rotation error.

$$\omega = \frac{2\pi}{T} = 2\pi f, \qquad 120 \text{ rpm} = 4\pi \text{ rad/s}$$

4 ## A worked wheel

A wheel of radius 0.35 m spins at 300 rpm. Find omega, the rim speed, the period, and the angle turned in 3 s.

$$\omega = 300(2\pi)/60 = 31.4 \text{ rad/s}, \qquad v = 11.0 \text{ m/s}$$

$$T = 0.20 \text{ s}, \qquad \theta = 94.2 \text{ rad} = 15 \text{ turns}$$

→ the rim moves at 40 km/h while the hub is at rest

DEFINE

the three quantities

$$\theta, \quad \omega = \dot{\theta}, \quad \alpha = \dot{\omega}$$

all shared by every point of a rigid rotator

NUMBERS

a check

R = 0.5 m at 120 rpm

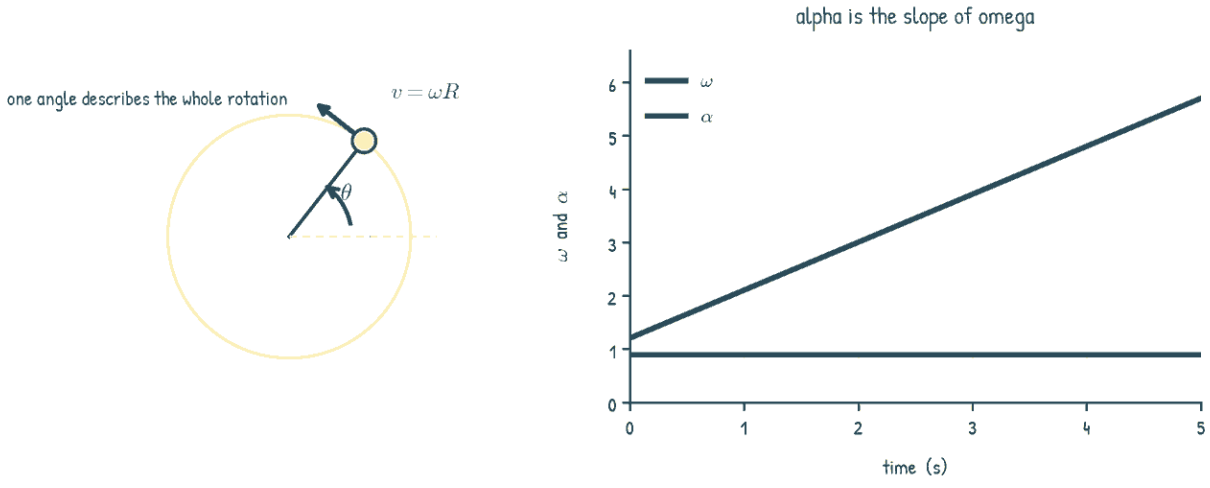
omega = 12.57 rad/s, v = 6.28 m/s, T = 0.5 s

TRAP

rpm into v = R omega

Convert first: radians per second, not revolutions per minute. One factor of 2 pi and one of 60.

omega_alpha



KEEP / every point shares the angle; only the radius converts it into a distance

What are the rotation equations for constant angular acceleration?

? A flywheel spins up from rest at a constant 3 rad/s². How many turns does it make in 4 s?

1

The same three equations

Every constant-acceleration result from one-dimensional motion has a rotational twin, because the derivation never used the fact that x was a length. Replace position by angle, velocity by angular velocity, acceleration by angular acceleration.

$$\omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \tfrac{1}{2}\alpha t^2, \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

2

Use the one without t when t is absent

The third equation is the workhorse of rotation questions: it connects speed, angle and acceleration without the clock. If the question gives an angle and asks for a final angular speed, that is the equation.

$$\omega^2 = \omega_0^2 + 2\alpha\theta \quad (\text{no } t)$$

3

Count turns, not radians

Questions often ask for revolutions. Convert at the end, dividing the angle by 2 pi, and check the answer is sensible: a wheel at 3 rad/s² from rest makes about half a turn per second squared.

→ theta in radians for every formula, turns only in the final sentence

4

A worked spin-up

A disc accelerates uniformly from 4 rad/s to 20 rad/s in 8 s. Find alpha, the angle turned, and the turns.

$$\alpha = \frac{20 - 4}{8} = 2.0 \text{ rad/s}^2$$

$$\theta = \frac{20^2 - 4^2}{2(2)} = 96 \text{ rad} = 15.3 \text{ turns}$$

→ the no-time equation was the shorter route: 96 rad directly

KEY RESULT

the trio

$$\omega = \omega_0 + \alpha t$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

NUMBERS

the opening question

alpha = 3 rad/s², t = 4 s

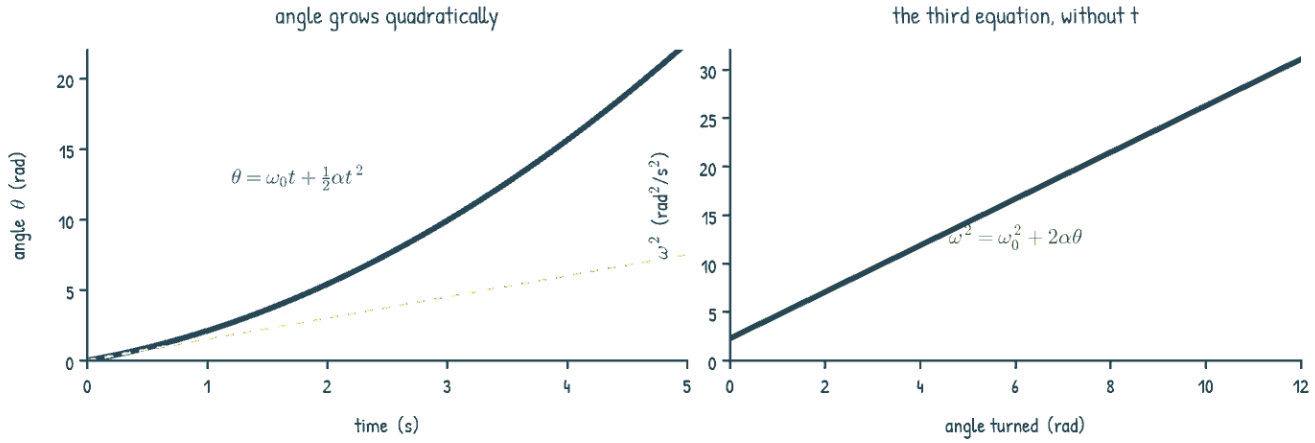
theta = 24 rad = 3.8 turns, omega = 12 rad/s

TRAP

alpha is not constant here

These three are for constant alpha only. If alpha varies, integrate it (see OLY 16).

alpha_kinematics



KEEP / the linear equations with x, v, a replaced by theta, omega, alpha

Why is angular velocity a vector?

? A wheel can spin about a vertical axis or a horizontal one, and the two are physically different. How does a single number capture that?

1 Direction along the axis

The scalar omega says how fast; the axis says about what. Put them together with the right-hand rule and angular velocity becomes a vector along the rotation axis. For a fixed axis that is overkill, but for a wheel that tilts it is the only correct bookkeeping.

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}, \qquad |\mathbf{v}| = \omega r \sin\phi$$

2 The cross product sorts out the geometry

For a point on a circle the radius and the axis are perpendicular, so the sine is one and the speed is omega r. For a point on the axis the cross product vanishes, so the axis itself does not move. That is exactly what a rigid rotation must do.

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} \quad \Rightarrow \quad v_\perp = \omega r_\perp$$

→ only the distance from the axis matters, not the distance from the origin

3 Adding angular velocities

Angular velocities about the same axis add as scalars; about different axes they add as vectors, and the order matters. That non-commutativity is why a spinning top precesses instead of simply falling, which is beyond this chapter but explains why the vector was needed.

4 A worked cross product

A disc spins at 12 rad/s about the z axis. Find the velocity of the point r = (0.4, 0.3, 0) m.

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = (0, 0, 12) \times (0.4, 0.3, 0) = (-3.6, 4.8, 0)$$

$$v = 6.0 \text{ m/s}, \qquad r_\perp = 0.5 \text{ m}$$

→ v = omega r_perp with the perpendicular distance, which is the check

DEFINE

the vector

$\boldsymbol{\omega}$ along the axis, $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$

NUMBERS

a check

omega = 4 rad/s about z, r = (0.3, 0, 0) m

v = (0, 1.2, 0) m/s

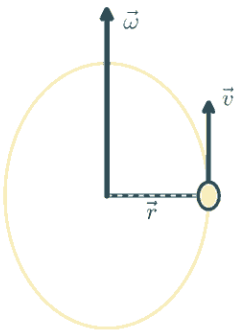
TRAP

using the distance from the origin

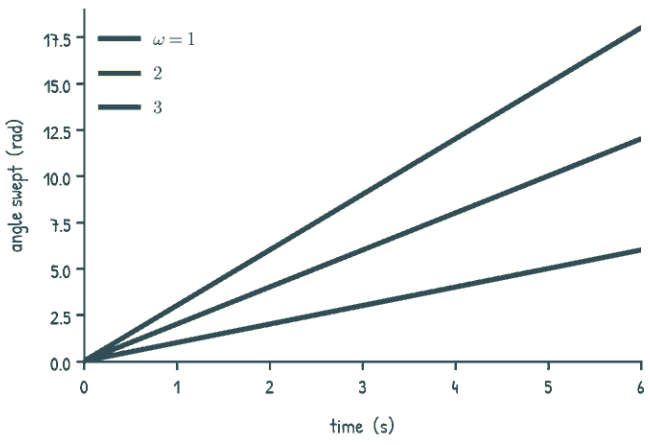
Use the perpendicular distance to the axis. A point on the axis has zero speed however far it is from the origin.

omega_vector

the right hand gives the direction along the axis



same axis, different rates



KEEP / omega points along the axis, and v = omega cross r gives the tangential speed

Why is circular motion at constant speed still accelerated?

? The speedometer reads a constant 60 km/h on a roundabout. Is the car accelerating, and if so, in which direction?

1

Velocity is a vector

Constant speed means the magnitude is fixed, not the direction. On a circle the direction turns continuously, so the velocity vector changes every instant, and a changing velocity is an acceleration by definition.

→ the acceleration is not along the motion; it is across it

2

Derivation 1: differentiate the vectors

Write the position as R times the radial unit vector and differentiate twice. The radial unit vector rotates at omega, so its derivative points along the tangent, and the second derivative brings back a term pointing inward.

$$\mathbf{a} = -R\omega^2\hat{r} + R\dot{\omega}\hat{\theta}$$

3

Derivation 2: the velocity triangle

Draw the velocity at two nearby instants. Both have length v, and the angle between them is the angle turned. The change is v times that angle, pointing inward, so the acceleration magnitude is v omega.

$$a_c = \frac{v^2}{R} = \omega^2 R = v\omega$$

→ two independent routes, one result: that is the standard JEE proof requirement

4

A worked roundabout

A car takes a 60 m radius roundabout at 45 km/h. Find the centripetal acceleration and the friction coefficient needed.

$$v = 12.5 \text{ m/s}, \qquad a_c = \frac{156}{60} = 2.60 \text{ m/s}^2$$

$$\mu = \frac{a_c}{g} = 0.27$$

→ below a dry-road 0.7, above an icy 0.1: the number decides the story

KEY RESULT

centripetal acceleration

$$a_c = \frac{v^2}{R} = \omega^2 R$$

directed toward the centre, at every instant

NUMBERS

a check

90 km/h = 25 m/s on R = 100 m

a_c = 6.25 m/s², about 0.64 g

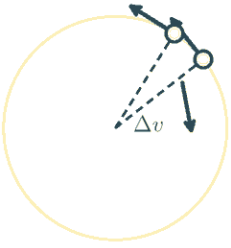
TRAP

calling it a force

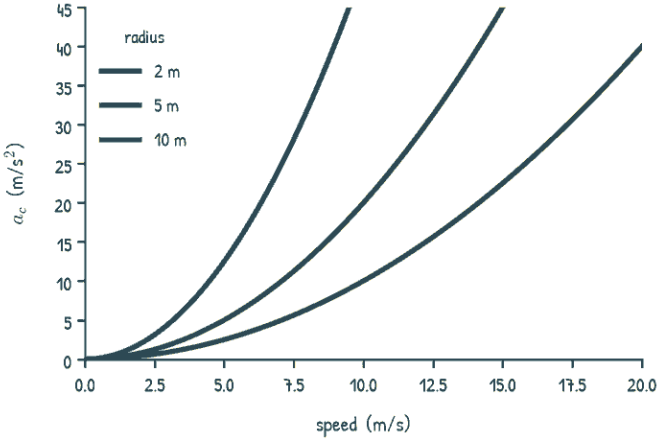
Centripetal acceleration is a kinematic fact. The force that supplies it is whatever happens to point inward: friction, tension, gravity, normal.

centripetal_two_ways

Δv points at the centre for any small turn



$a_c = v^2/R$: speed is squared, radius divides



KEEP / the speed is constant but the direction turns, so the velocity changes toward the centre

How much of the energy is actually available?

? Two blocks collide. The total kinetic energy is 100 J, but only part of it can become heat or deformation. Which part?

1 ## Two pieces of one number

Write each velocity as the centre-of-mass velocity plus a velocity relative to it. The cross terms cancel because the total momentum in the CM frame is zero, and the kinetic energy separates into two independent pieces.

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + K_{\text{rel}}, \qquad M = m_1 + m_2$$

2 ## The internal part is the reducible one

No internal force can change the CM velocity, so the drift energy is untouchable from inside. Everything a collision, an explosion or a spring can convert comes from the relative part, and that is why the reduced mass appears in every collision formula.

$$K_{\text{rel}} = \tfrac{1}{2}\mu v_{\text{rel}}^2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

→ a perfectly inelastic collision converts exactly K_rel into heat

3 ## The two limits to remember

Equal masses moving toward each other at the same speed: the CM is at rest and all the energy is internal, so all of it can be lost. A light particle hitting a heavy one at rest: almost all the energy is drift, so very little can be converted, which is why a bullet heats a wall far more easily than a ping-pong ball heats a block.

$$m_1 = m_2 : \; K_{\text{rel}} = \tfrac{1}{2}K_{\text{tot}}, \qquad m_1 \ll m_2 : \; K_{\text{rel}} \approx \tfrac{m_1}{m_2}K_{\text{tot}}$$

4 ## A worked split

A 3 kg block at 4 m/s hits a 1 kg block at rest and they stick. Find the CM speed, the CM energy, and the energy lost.

$$V = \frac{12}{4} = 3 \text{ m/s}, \qquad K_{\text{cm}} = \tfrac{1}{2}(4)(9) = 18 \text{ J}$$

$$K_i = 24 \text{ J} \;\Rightarrow\; \text{lost } 6 \text{ J}, \;\; \text{which is } K_{\text{rel}}$$

→ the 18 J of drift survives the collision untouched

KEY RESULT

the decomposition

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + \tfrac{1}{2}\mu v_{\text{rel}}^2$$

NUMBERS

a check

2 kg at 3 m/s meets 1 kg at rest

mu = 2/3, K_rel = 3 J of the 9 J total

INSIGHT

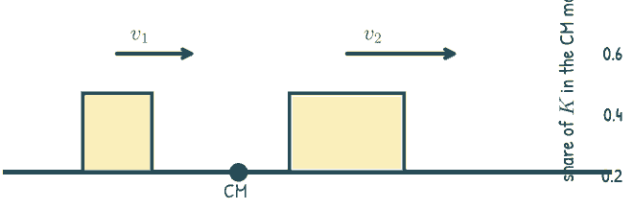
why reduced mass appears

It is the mass that carries the internal motion. Every two-body collision formula is a one-body formula with mu in place of m.

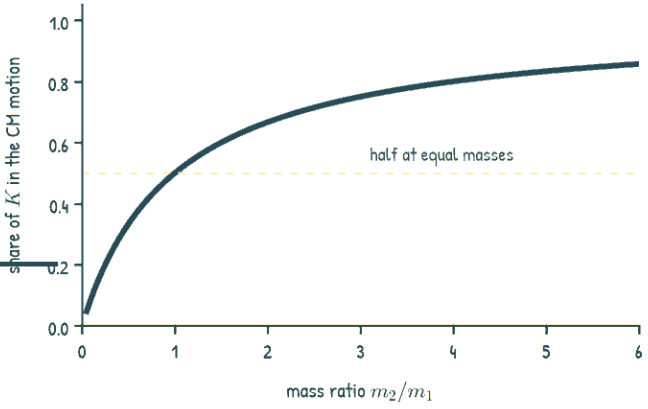
cm_decomposition

split the motion into drift plus internal motion

$$K = \tfrac{1}{2}Mv_{\text{cm}}^2 + \tfrac{1}{2}\mu v_{\text{rel}}^2$$



the drift usually carries most of it



KEEP / split the kinetic energy into drift of the centre of mass plus internal motion

Why does angular momentum keep a planet out of the Sun?

? Gravity pulls inward with nothing to oppose it. Why does a planet not simply fall in?

1 ## Separate the two motions

In a central field the angular momentum is constant, so the angular part of the kinetic energy is fixed by the radius alone. Move it to the other side of the energy equation and it behaves like an extra potential.

$$E = \tfrac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r), \qquad U_{\text{eff}} = U(r) + \frac{L^2}{2mr^2}$$

2 ## Read the effective curve

The added term is positive and falls as 1/r squared, so it dominates at small radii and creates a minimum when combined with an attractive potential. The minimum is the circular orbit; a small radial oscillation around it is an ellipse.

$$\frac{dU_{\text{eff}}}{dr} = 0 \;\Rightarrow\; r_0 = \frac{L^2}{GMm^2}$$

→ no force pushes outward: the barrier is the cost of conserving angular momentum

3 ## Where it is genuinely useful

The same construction gives the critical angular momentum below which a particle falls in, the turning points of a comet, and the stability condition for a circular orbit. It also explains the bead on a rotating hoop (OLY 08), which is the same trick in a rotating frame.

4 ## A worked barrier

A satellite orbits at 7000 km from the Earth centre. Find the effective potential at that radius per unit mass, and compare the two terms.

$$-\frac{GM}{r} = -5.69 \times 10^{7}\ \text{J/kg}$$

$$+\frac{v^2}{2} = +2.85 \times 10^{7}\ \text{J/kg}$$

→ the sum is minus half the magnitude: the virial relation for a circular orbit

KEY RESULT

effective potential

$$U_{\text{eff}} = U(r) + \frac{L^2}{2mr^2}$$

the angular part of K, rewritten as a barrier

NUMBERS

a check

Earth: L = 2.66e40, GMm² gives r = 1.5e11 m

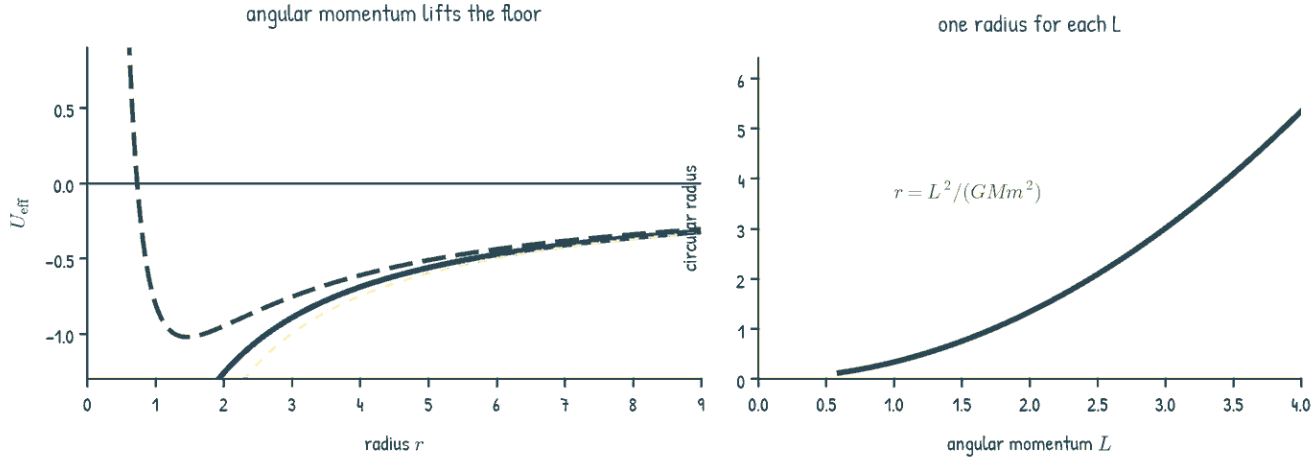
the minimum of U_eff reproduces the orbit radius

TRAP

calling it a centrifugal force

In an inertial frame there is no outward force. The barrier is energy bookkeeping; the only force is inward.

effective_potential



KEEP / the angular term is a repulsive barrier made of kinetic energy, not a force

How far does a bouncing ball travel before it stops?

? A ball is dropped from h and returns to e²h after each bounce. The bounces are infinite; is the total distance infinite too?

1

Write the two series

The first fall is h. Every later bounce contributes a rise and a fall, each shorter by e squared. The flight time scales by e rather than e squared, because time goes as the square root of height.

$$s = h\frac{1+e^2}{1-e^2}, \qquad t = \sqrt{\frac{2h}{g}} \cdot \frac{1+e}{1-e}$$

2

The energy view

Each impact multiplies the speed by e and therefore the kinetic energy by e squared. The energy lost per bounce is a fixed fraction of what is left, so the sequence is geometric and the total lost before rest is exactly the initial mgh.

$$E_n = e^{2n}mgh, \qquad \sum E_{\text{lost}} = mgh$$

→ the ball never gets negative energy: the sum is bounded by the initial store

3

Why the time is finite while the bounces are not

The bounce count diverges but the durations shrink faster than the count grows. The series converges to a finite time, so the ball comes to rest in finite time after infinitely many bounces, which is the standard counterexample to "infinitely many events need infinite time".

$$t_0(1 + 2e + 2e^2 + \ \dots\ ) = t_0\frac{1+e}{1-e}$$

4

A worked bounce

A ball dropped from 1.8 m rebounds to 1.15 m. Find e, the total distance and the total time.

$$e = \sqrt{1.15/1.8} = 0.80, \qquad s = 1.8\frac{1.64}{0.36} = 8.2\ \text{m}$$

$$t = \sqrt{0.367} \cdot 9 = 0.606(9) = 5.45\ \text{s}$$

→ one bounce is over in 0.6 s; the infinite tail takes five more

KEY RESULT

the closed forms

$$s = h\frac{1+e^2}{1-e^2}, \qquad t = \sqrt{\frac{2h}{g}} \cdot \frac{1+e}{1-e}$$

NUMBERS

a check

h = 10 m, e = 0.5

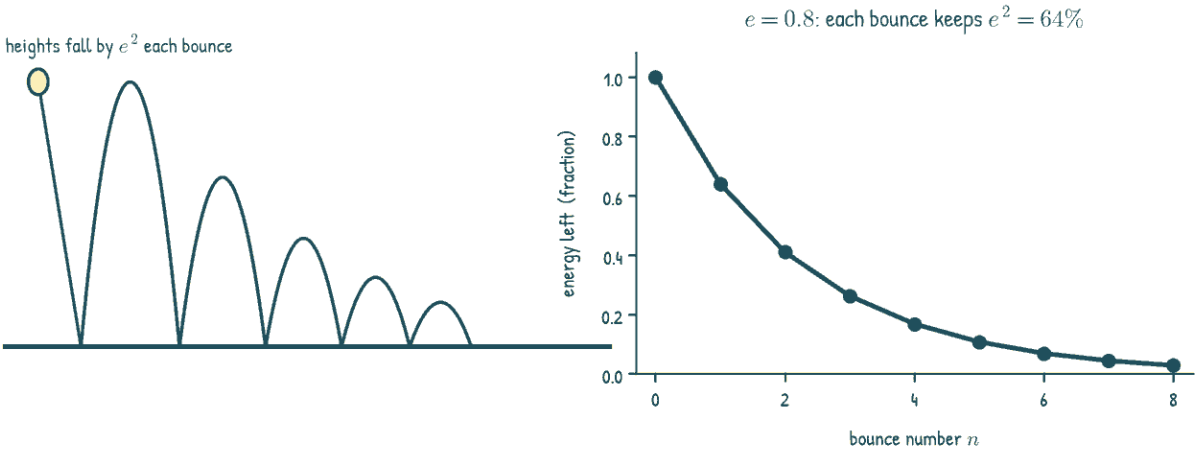
s = 16.7 m, t = 4.29 s

TRAP

the factor of two

Every bounce after the first contributes twice its height, once up and once down.

bounce_energy



KEEP / two geometric series, one for distance and one for time, both with finite sums

Why does a rolling ball beat a sliding block downhill?

? A ball and a block are released from the same height on the same slope. The block has no friction, the ball rolls. Which arrives first?

1 Two kinetic terms

A rolling body translates and rotates at once. With the rolling condition the rotation is fixed by the speed, so the total kinetic energy is a fixed multiple of half m v squared.

$$K = \tfrac{1}{2}mv^2 + \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}mv^2\left(1 + \tfrac{I}{mR^2}\right)$$

2 The shape sets the multiple

A ring has I = mR squared, so it carries twice the energy of a sliding block at the same speed and is the slowest. A disc gives three quarters, a solid sphere seven tenths. The order of arrival downhill is always sphere, disc, ring, regardless of mass and radius.

$$K:\ \tfrac{1}{2}mv^2\ (\text{slide}),\ \tfrac{3}{4}mv^2\ (\text{disc}),\ \tfrac{7}{10}mv^2\ (\text{sphere}),\ mv^2\ (\text{ring})$$

→ the ratio I/mR squared is the only thing that matters

3 Where the energy goes on a slope

Energy conservation from height h gives the speed at the bottom with the same factor in the denominator. Static friction does the work that spins the body up, but it does no net work on the rolling body because the contact point does not slip.

$$v^2 = \frac{2gh}{1 + I/(mR^2)}$$

4 A worked roll

A solid sphere, a disc and a ring roll from rest down a 2 m high slope. Find each bottom speed.

$$v = \sqrt{\frac{2gh}{1 + I/mR^2}}:\ 5.29,\ 5.11,\ 4.43\ \text{m/s}$$

$$\text{sphere}\ (2/5),\ \text{disc}\ (1/2),\ \text{ring}\ (1)$$

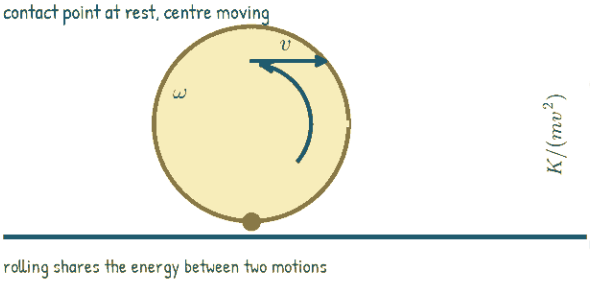
→ all three take the same time only if the slope is long enough to reach these speeds

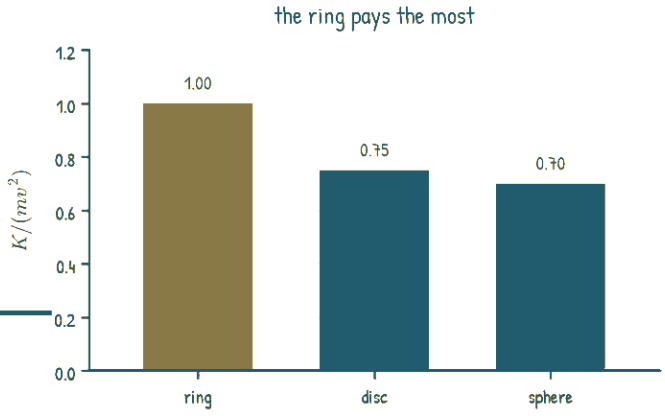
KEY RESULT
rolling energy
$K = \tfrac{1}{2}mv^2\left(1 + \tfrac{I}{mR^2}\right)$

NUMBERS
a check
solid sphere from 1.4 m
v = sqrt(2 x 9.8 x 1.4 / 1.4) = 4.43 m/s

TRAP
adding friction work
Static friction at a non-slipping contact does no work on the rolling body. Do not subtract f d from its energy.

rolling_energy





KEEP / a rolling body spends part of its energy on rotation, so it descends more slowly

Why is the block on a moving wedge slower than sqrt(2gh)?

? A block slides down a smooth wedge that is free to slide on a smooth table. The textbook answer sqrt(2gh) must be wrong. Why?

1

Two unknowns, two equations

Horizontal momentum is conserved for the pair, and mechanical energy is conserved for the pair. Those two equations determine the wedge speed and the relative sliding speed; the block ground speed then follows from the vector sum.

$$u^2 = \frac{2gh(M+m)}{M+m\sin^2\theta}, \qquad V = \frac{mu\cos\theta}{M+m}$$

2

The error the source warns about

Treating u as the block speed is the standard mistake. u is measured along the face relative to the wedge; the ground speed is the vector sum of u and V, and is smaller for a finite wedge.

$$v_b^2 = u^2\left(\sin^2\theta + \left(\frac{M}{M+m}\right)^2\cos^2\theta\right)$$

→ the block ground speed is below sqrt(2gh) for every finite M

3

The two limits

An infinitely heavy wedge cannot recoil and the block recovers sqrt(2gh). A massless wedge carries almost no energy, so the block again approaches the same speed. The deficit peaks when the masses are comparable.

$$M\to\infty:\ v_b\to\sqrt{2gh}, \qquad M\to 0:\ v_b\to\sqrt{2gh}$$

4

A worked wedge

m = 1 kg, M = 4 kg, theta = 30°, released from 0.5 m up the face. Find both speeds at the bottom of the face.

$$u^2 = \frac{2(9.8)(0.5)(5)}{4.25} = 11.5 \ \Rightarrow\ u = 3.39\ \text{m/s}$$

$$V = \frac{1(3.39)(0.866)}{5} = 0.59\ \text{m/s}$$

→ the block drops 0.5 m but its ground speed is only 2.9 m/s, below the 3.1 m/s fixed-wedge value

KEY RESULT

the pair

$$mgh = \tfrac{1}{2}MV^2 + \tfrac{1}{2}mv_b^2$$

plus horizontal momentum conservation

NUMBERS

a check

m = 1, M = 4, theta = 30°, h = 0.5 m

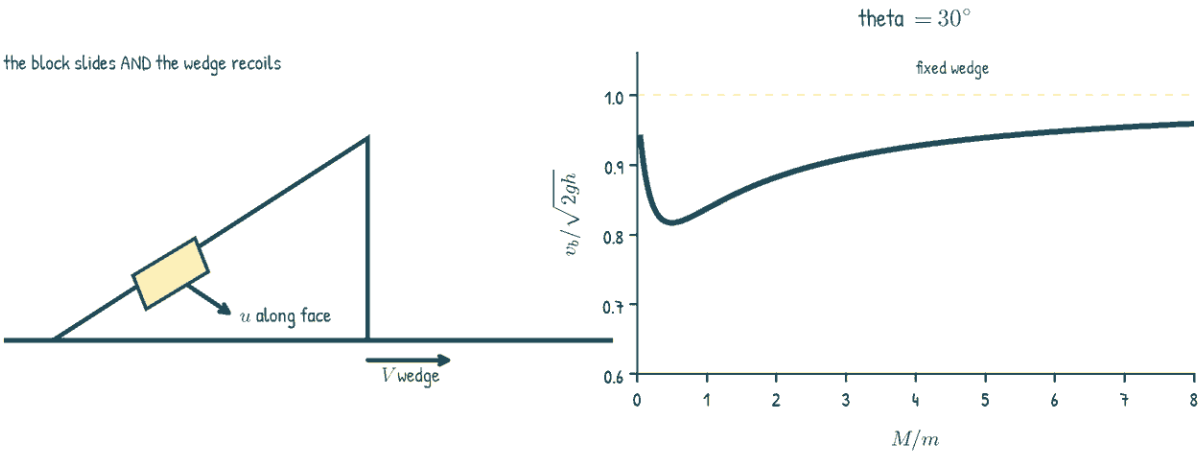
u = 1.60 m/s, V = 0.28 m/s, v_b = 1.15 m/s

TRAP

one-body energy

The block alone does not conserve energy: the normal force does work on it because the face moves.

wedge_energy



KEEP / the wedge takes a share of the energy, so the block cannot have all of it

How much energy does it take to reach orbit?

? Escape needs the full well depth. Orbit needs less, but not as little as you might hope. What fraction?

1

The two energies

The energy at rest on the surface is minus GMm over R. The energy of a circular orbit is minus GMm over twice r, which is the virial relation for a circular orbit. The difference is what the rocket must supply.

$$E_{\text{orbit}} - E_{\text{surface}} = GMm\left(\tfrac{1}{R} - \tfrac{1}{2r}\right)$$

2

The ratio that surprises

For a low orbit, r is close to R and the bracket is about one over twice R. The escape energy is GMm over R. So orbit costs half of escape, and the remaining half is the kinetic energy the satellite keeps in orbit, not a loss.

$$\frac{\Delta E_{\text{orbit}}}{\Delta E_{\text{escape}}} = \frac{1}{2} - \frac{R}{2r}$$

→ the orbital speed is 1/sqrt(2) of the escape speed, which is the same statement

3

Why it matters

Reaching orbit is not mainly about height; it is mostly about the horizontal speed. A sounding rocket that reaches 400 km with no horizontal speed has done far less than a satellite at the same height, which is why launch vehicles pitch over early.

4

A worked launch

Find the energy to lift 1 kg from the Earth surface to a 400 km orbit, and compare it with escape.

$$\Delta E = GM\left(\tfrac{1}{R} - \tfrac{1}{2r}\right) = 6.26 \times 10^7 - 2.94 \times 10^7$$

$$= 3.3 \times 10^7 \ \text{J/kg} = 33 \ \text{MJ/kg}$$

→ escape from the same point is 62.5 MJ/kg, so orbit costs just over half

KEY RESULT

the ratio

$$\Delta E_{\text{orbit}} = \tfrac{GMm}{2R} \ \text{(low orbit)}, \qquad = \tfrac{1}{2}\Delta E_{\text{esc}}$$

NUMBERS

a check

Earth: v_esc = 11.2 km/s, v_orbit = 7.9 km/s

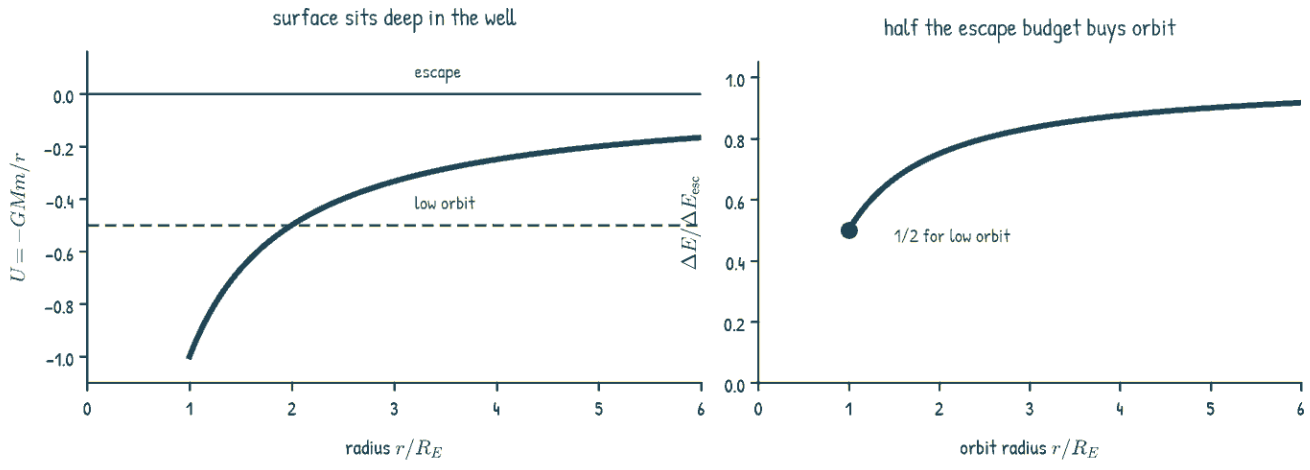
ratio of energies: 1/sqrt(2) squared = 0.5

TRAP

counting the orbital KE as wasted

It is stored, not lost: bringing the satellite down returns it. Only the drag losses are gone.

launch_energy



KEEP / low orbit costs half the escape energy, and the other half is the speed you keep

Where does the missing half of the work go?

? A chain is pulled off a table at constant speed v. The power delivered is F v, but the energy in the chain grows more slowly. Where is the difference?

1 ## The two rates

The force must support the hanging weight and supply the momentum flux. Multiply by v to get the power delivered, and compare it with the rate of change of the mechanical energy.

$$P = Fv = \lambda x g v + \lambda v^3, \qquad \dot{E} = \lambda x g v + \tfrac{1}{2}\lambda v^3$$

2 ## The difference is the collision

Each link is at rest and then shares the speed v of a moving system. That is a perfectly inelastic capture, and half the energy invested in the capture is unavoidably lost, exactly as in any inelastic collision.

$$\dot{Q} = \tfrac{1}{2}\lambda v^3$$

→ the loss scales as v cubed, so fast pulling is disproportionately wasteful

3 ## The same ledger as the rocket

Both the chain and the rocket are open systems: mass crosses the boundary, and the momentum flux must be included. The chain is the inelastic limit of the rocket, with the links being captured instead of expelled.

4 ## A worked unroll

A 4 kg, 2 m chain lies coiled on a table. It is pulled up at a steady 1.5 m/s. Find the pull and the loss rate when 1.2 m is up.

$$F = \lambda x g + \lambda v^2 = 2(9.8)(1.2) + 2(2.25) = 28.0 \text{ N}$$

$$\dot{Q} = \tfrac{1}{2}\lambda v^3 = 3.4 \text{ W of the } 42 \text{ W delivered}$$

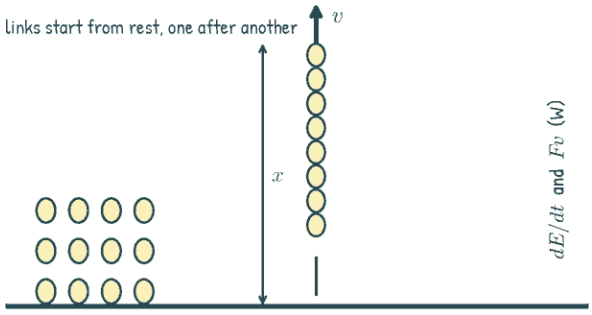
→ 8 per cent is lost in the jerk at this speed, and it grows as v cubed

KEY RESULT
the audit
$$F = \lambda x g + \lambda v^2, \qquad \dot{Q} = \tfrac{1}{2}\lambda v^3$$

NUMBERS
a check
lambda = 2 kg/m, x = 2, v = 1 m/s
F = 41.2 N, loss 1 W of the 41.2 W delivered

INSIGHT
the general rule
Mass captured by a moving system always costs a factor of two: half to motion, half to heat.

chain_unroll





KEEP / links are jerked into motion inelastically, and the lost half is heat

Why does a spinning hoop make the bead climb?

? A bead slides freely on a smooth hoop spinning about its vertical diameter. At high enough speed it leaves the bottom. Where does the energy come from?

1

Two contributions, one curve

Gravity favours the bottom and the rotation favours the equator, because the bead must be further from the axis to have a large radius at high angular speed. Adding the two produces an effective potential whose shape depends on omega.

$$U_{\text{eff}}(\theta) = -mgR\cos\theta - \tfrac{1}{2}m\omega^2R^2\sin^2\theta$$

2

The threshold

Setting the derivative to zero gives the bottom and, for fast enough rotation, a pair of side positions. The condition for the side positions to exist is that the rotational term beats gravity at the bottom.

$$\omega > \omega_c = \sqrt{\frac{g}{R}}, \qquad \cos\theta_0 = \frac{g}{\omega^2 R}$$

→ below the critical speed the only stable seat is the bottom

3

Where the energy comes from

The hoop is driven, so it does work on the bead through the normal force, which has a component along the bead motion once the hoop is rotating. Energy is supplied by the motor, and the bead settles where the effective potential is lowest.

4

A worked hoop

A bead on a 0.4 m hoop which spins at 5 rad/s. Find the critical speed and the equilibrium angle.

$$\omega_c = \sqrt{9.8/0.4} = 4.95 \text{ rad/s}$$

$$\cos\theta = \frac{9.8}{25(0.4)} = 0.98 \;\Rightarrow\; \theta = 11^\circ$$

→ just above the threshold the bead sits very close to the bottom

KEY RESULT

the result

$$\omega_c = \sqrt{g/R}, \qquad \cos\theta_0 = g/(\omega^2 R)$$

NUMBERS

a check

R = 0.5 m, omega = 6 rad/s

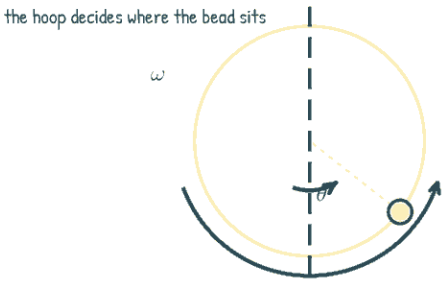
omega_c = 4.43, theta = 57°

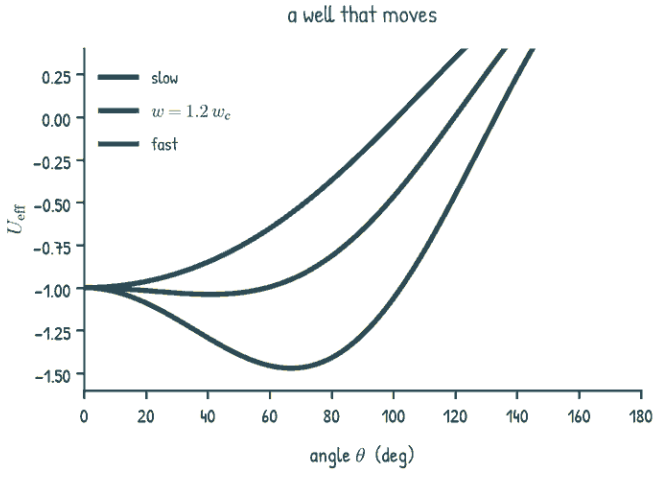
TRAP

energy conservation in the lab frame

The bead alone does not conserve energy. Use the effective potential, or include the work done by the hoop.

bead_energy





KEEP / the hoop does work on the bead, and in the rotating frame that is one effective potential

What does the coefficient of restitution actually measure?

? Two balls drop from the same height onto the same floor and bounce to different heights. What single number separates them?

1 Define it on speeds

The coefficient of restitution is the ratio of separation speed to approach speed along the normal. It is a property of the pair of surfaces and the impact speed, not of the mass.

$$e = \frac{v_{\text{sep}}}{v_{\text{app}}}, \qquad 0 \le e \le 1$$

2 What it does to energy

Since energy goes as speed squared, one bounce multiplies the mechanical energy by e squared. That is why the rebound height is e squared times the drop height, and why a bouncy ball is specified by its rebound ratio rather than by a force.

$$\frac{h_{n+1}}{h_n} = e^2, \qquad \frac{K_{n+1}}{K_n} = e^2$$

→ e is a ratio of speeds; the energy ratio is its square

3 The two extremes

e = 1 is elastic: no energy lost, the ball returns to the same height for ever. e = 0 is perfectly inelastic: the ball stays on the floor, and all the kinetic energy becomes heat and sound. Real balls sit between 0.6 and 0.9.

→ a superball at e = 0.9 keeps 81 per cent of its energy per bounce

4 A worked restitution

A ball falls 3 m and rebounds to 1.08 m. Find e and the fraction of energy kept.

$$e = \sqrt{1.08/3} = 0.60$$

$$\frac{K_f}{K_i} = e^2 = 0.36$$

→ it loses nearly two thirds of its energy at one impact

DEFINE

restitution

$e = v_{\text{sep}}/v_{\text{app}}$

no mass, no height: a property of the impact

NUMBERS

a check

drops 2 m, rebounds 1.28 m

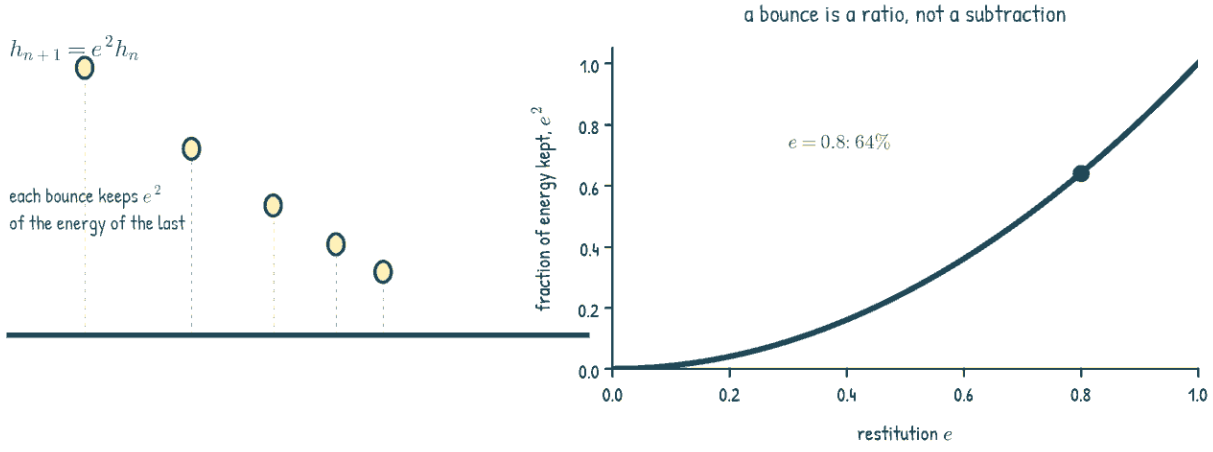
e = sqrt(1.28/2) = 0.8, energy kept 64 per cent

TRAP

e as an energy ratio

It is not. Energy scales as its square, so a rebound to half the height means e = 0.71, not 0.5.

restitution_energy



KEEP / e is the ratio of relative speeds, so energy is multiplied by e squared per impact

Who pays for the kinetic energy of a flywheel?

? A motor of constant power spins up a flywheel. The angular speed grows, but not linearly. Why, and who supplies the difference?

1 ## Power into rotation

The rotational kinetic energy is half I omega squared, so its derivative gives the power. That is the rotational twin of P = F v, with torque in place of force and omega in place of speed.

$$P = \frac{d}{dt}\left(\tfrac{1}{2} I\omega^2\right) = I\omega\dot{\omega} = \tau\omega$$

2 ## Integrate

With P constant the differential equation separates and gives omega proportional to the square root of time. The torque falls as one over omega, which is why a constant-power motor is slow at the start and cannot spin a wheel up evenly.

$$\omega = \sqrt{\frac{2Pt}{I}}, \qquad \tau = \frac{P}{\omega}$$

→ the same structure as v = sqrt(2Pt/m) for a car

3 ## Where the energy sits

All the work done by the motor ends up in the wheel: none is lost if the bearing is ideal. The question in the title is answered by the motor, and the only subtlety is the non-uniform rate at which it is delivered.

4 ## A worked flywheel

A flywheel of I = 8 kg m² is spun by a 300 W motor from rest. Find omega at 6 s and the time to 40 rad/s.

$$\omega = \sqrt{2(300)(6)/8} = 21.2 \text{ rad/s}$$

$$t = \frac{8(1600)}{2(300)} = 21.3 \text{ s}$$

→ the last 20 rad/s takes three times as long as the first 20

KEY RESULT

constant-power spin-up

$$\omega = \sqrt{2Pt/I}$$

and torque = P/omega

NUMBERS

a check

I = 12 kg m², P = 500 W

omega(10 s) = sqrt(833) = 28.9 rad/s

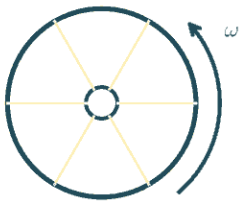
TRAP

using tau = I alpha with constant tau

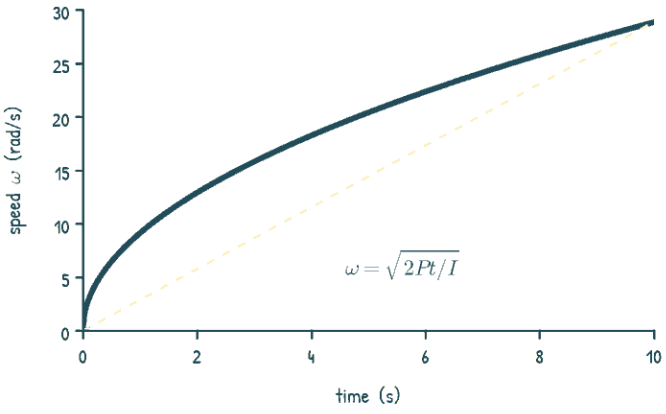
Constant power is not constant torque. Set tau = P/omega and solve the differential equation.

flywheel

a constant-power motor cannot spin up a wheel linearly



power divided by torque is angular speed



KEEP / at constant power the torque falls as one over omega, so the spin-up slows as root t

Can a force that is not there do work?

? A ball hangs in an accelerating cart. In the cart frame it swings backward and gains speed. Which force did the work?

1 ## Two frames, two accounts

In the ground frame the tension does the work that changes the ball speed; the pseudo force does not exist. In the cart frame the ball moves backward along the pseudo force, which does positive work equal to m a times that displacement.

$$W_{\text{pseudo}} = \int (-m\mathbf{a}_0) \cdot d\mathbf{r}$$

2 ## Why the frame version still balances

In a uniformly accelerating frame the pseudo force is constant and uniform, so it has a potential m a dot r and the energy book closes. In a rotating frame the pseudo forces include Coriolis, which is velocity-dependent and does no work, and centrifugal, which does and has a potential.

$$U_{\text{cf}} = -\tfrac{1}{2} m\omega^2 r_\perp^2$$

→ the bead on the hoop is the standard consumer of that potential

3 ## The exam rule

You may use energy in an accelerating frame only if you include the pseudo potential. Most students forget it and get a speed that is too large. The safe route is to compute energy in the inertial frame and forces in whichever frame is easier.

4 ## A worked frame

A 0.2 kg ball hangs in a cart accelerating at 4 m/s². It swings 0.25 m backward relative to the cart. Find the pseudo work and the ball speed in the cart frame.

$$W_{\text{pseudo}} = m a_0 d = 0.2(4)(0.25) = 0.20 \text{ J}$$

$$v = \sqrt{2(0.20)/0.2} = 1.41 \text{ m/s}$$

→ in the ground frame the tension did that work instead

KEY RESULT

pseudo potential

$$U_{\text{pseudo}} = -m\mathbf{a}_0 \cdot \mathbf{r}$$

centrifugal: -half m omega squared r squared

TRAP

Coriolis in an energy equation

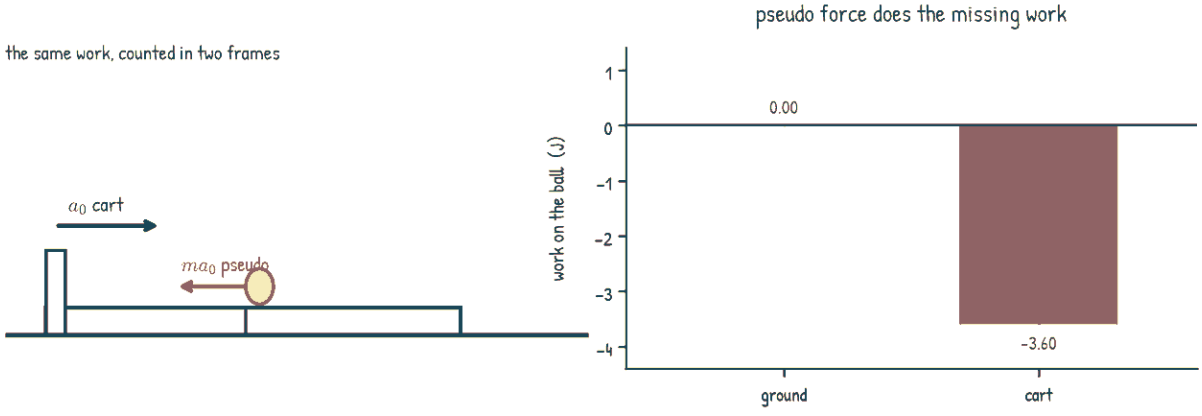
Coriolis is perpendicular to the velocity, so it does no work and has no potential. Only the uniform and centrifugal parts enter energy.

INSIGHT

the safe habit

Energy in the inertial frame, forces in the convenient frame. Mixing the two is where signs go wrong.

pseudo_work



KEEP / in an accelerating frame the pseudo force does work, and the frame energy book stays closed

Is one closed loop enough to prove a force is conservative?

? You are handed a force field and one closed loop. If the loop integral vanishes, may you conclude that a potential exists?

1 ## What the definition demands

Conservatism is a statement about all paths between any two points. A single loop that happens to close is a single data point; the field could still fail elsewhere.

$$\oint \mathbf{F} \cdot d\mathbf{r} = 0 \text{ for every loop}$$

2 ## The local test

For a force field in three dimensions the loop condition is equivalent to the curl vanishing everywhere in a simply connected region. In two dimensions that is the single condition that the cross derivative matches.

$$\nabla \times \mathbf{F} = 0 \quad \Longleftrightarrow \quad \partial_x F_y = \partial_y F_x$$

→ a vortex field passes no loop, and its curl is not zero

3 ## Why this matters in mechanics

Every potential energy you use in this chapter was licensed by that test. Gravity, the spring and the uniform field pass; friction and drag fail because they depend on the direction of travel or on speed, so no function of position can reproduce them.

4 ## A worked curl

Test F = (2xy, x²) N for conservatism, and find its potential.

$$\partial_x F_y = 2x = \partial_y F_x \text{ so the curl vanishes}$$

$$U = -x^2 y + \text{const}$$

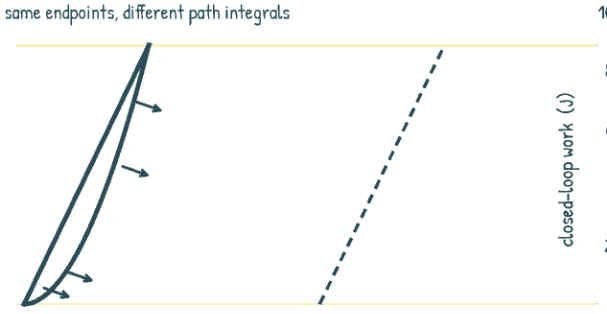
→ a vanishing curl plus a simply connected region is the licence for a potential

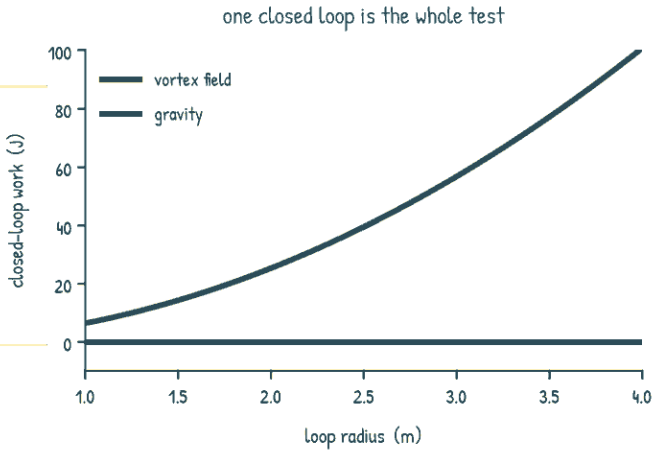
GATE
the two tests
global: all loop integrals zero. local: curl zero in a simply connected region.

NUMBERS
a check
F = (−y, x): curl = 2, loop work = 2 x area
nonzero for every loop: not conservative

TRAP
one loop as proof
A vanishing loop integral on a convenient circle proves nothing about the rest of the plane.

path_integral





KEEP / one loop proves nothing; every loop must vanish, which is what the curl tests

Why does doubling speed need eight times the power?

? A cyclist at 20 km/h is comfortable. At 40 km/h the same rider is exhausted. Why is the cost not twice?

1 ## The force law

At everyday speeds the drag force is proportional to the square of the speed. The power needed to overcome it is force times speed, so it goes as the cube.

$$f = bv^2, \qquad P = bv^3$$

2 ## The top speed

At constant speed the drive equals the drag, so a fixed power sets a maximum speed proportional to the cube root of the power. Four times the power buys only 1.6 times the speed, which is why the land speed record is so expensive.

$$v_{\max} = \left(\frac{P}{b}\right)^{1/3}$$

→ the cube root is the sting: 100 kW for a car means about 55 m/s with b = 0.6

3 ## The approach to the top

With a constant driving force the speed approaches terminal speed exponentially, and with constant power it approaches the ceiling from below without ever reaching it. Either way the last few metres per second cost most of the time.

$$v(t) = v_\infty(1 - e^{-t/\tau}) \quad \text{(constant drive)}$$

4 ## A worked drag

A cyclist and bike of 85 kg push 250 W. With b = 0.32, find the top speed on the flat.

$$250 = 0.32v^3 \;\Rightarrow\; v = (781)^{1/3} = 9.2 \text{ m/s}$$

$$\text{at } 500 \text{ W}: \; v = 11.6 \text{ m/s}$$

→ twice the power buys 26 per cent more speed, which is the whole story of cycling

KEY RESULT

drag power

$$f = bv^2, \qquad P = bv^3$$

NUMBERS

a check

b = 0.6, P = 100 kW

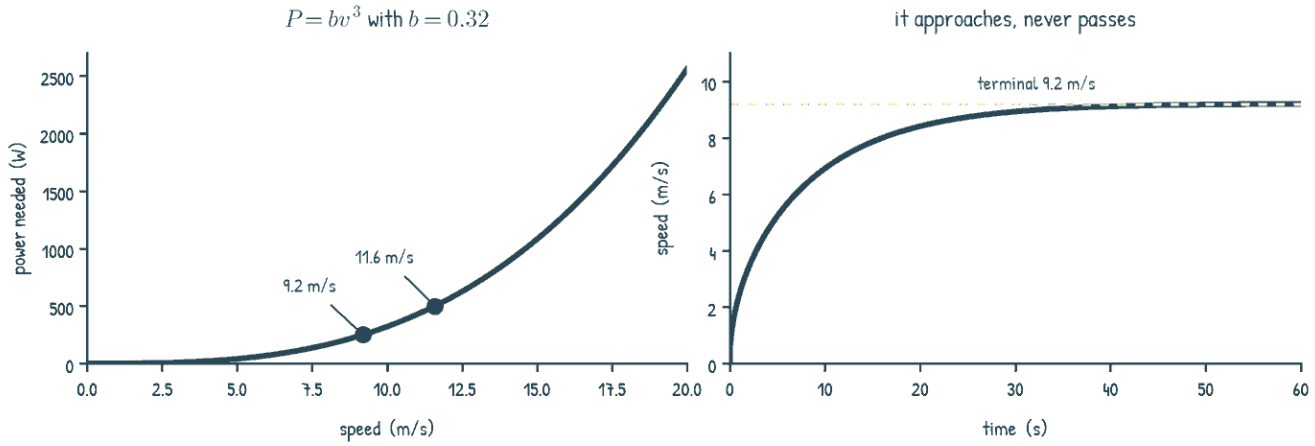
v_max = (100000/0.6)^(1/3) = 55 m/s

TRAP

linear scaling

Doubling speed quadruples the force and multiplies the power by eight. Linear intuition is wrong here by a factor of eight.

drag_power



KEEP / air resistance grows as the square of speed, and power as its cube

Where does the energy bookkeeping stop working?

? The chapter treats mechanical energy as exact. At what speed, and at what scale, does the ledger need new terms?

1 **Two fences**

Special relativity replaces half m v squared by (gamma minus one) m c squared once v is a noticeable fraction of c. Quantum mechanics makes the energy spectrum discrete once the action is comparable to h bar. Inside both fences the classical ledger is exact.

$$K = (\gamma - 1)mc^2 \approx \tfrac{1}{2}mv^2 \text{ for } v \ll c$$

2 **What survives**

The work-energy theorem survives both revolutions: work still equals the change in kinetic energy, with the relativistic kinetic energy. What does not survive is the formula half m v squared, and with it every classical estimate above about a tenth of the speed of light.

$$W = \Delta K \text{ always;} \quad K = \tfrac{1}{2}mv^2 \text{ only at low speed}$$

→ the theorem is more general than the formula it was derived from

3 **The practical boundary**

For a car, a satellite, a bullet, even a spacecraft at 10 km/s, gamma differs from one by less than one part in ten million. The relativistic correction is worth computing for electrons, for particle accelerators, and for nothing in this chapter.

4 **A worked ratio**

An electron at 2.0e8 m/s: what fraction of its kinetic energy does the classical formula miss?

$$\gamma = 1.34, \qquad K = (\gamma - 1)mc^2 = 0.342(511) = 175 \text{ keV}$$

$$\tfrac{1}{2}mv^2 = 114 \text{ keV} \;:\; \text{classical is } 35\ \%\ \text{low}$$

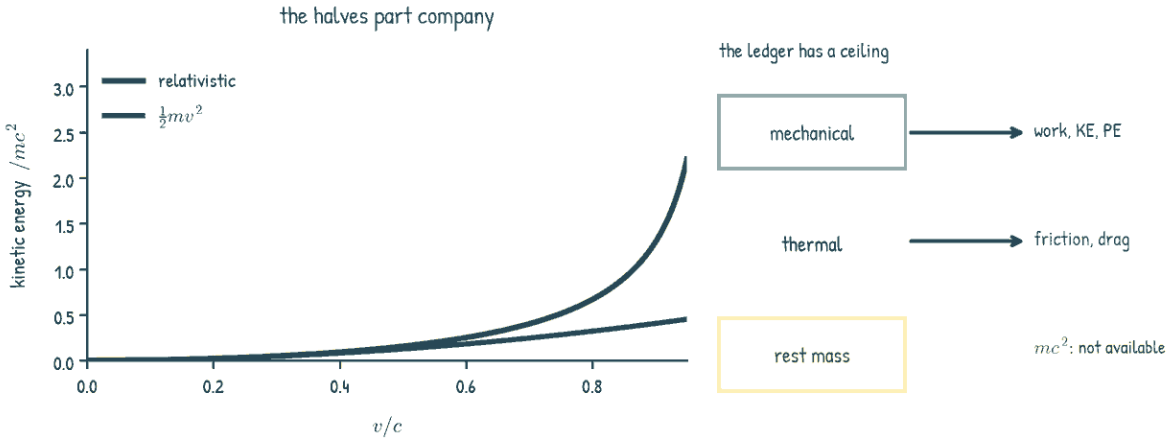
→ at v/c = 0.67 the classical formula is no longer an approximation, it is wrong

GATE
the ratio to check
Compute v/c. Below 0.1, classical is safe to a fraction of a per cent.

TRAP
quoting five figures
With g given to two figures and friction coefficients guessed, four-figure energy answers are a claim the data cannot support.

INSIGHT
what the chapter really teaches
Not the formulas, but the habit: name the store, name the loss, and check the ledger balances.

energy_limits



KEEP / the classical ledger is a low-speed approximation, and the rest mass joins it at high speed

What if the speed changes as well as the direction?

? A car on a curve brakes while turning. What is the total acceleration, and at what angle does it point?

1 ## Split the acceleration

Resolve along the tangent and the normal. The tangential part is the rate of change of speed; the normal part is v squared over R. They are perpendicular, so the magnitudes add in quadrature.

$$a_t = \frac{dv}{dt}, \qquad a_n = \frac{v^2}{R}, \qquad a = \sqrt{a_t^2 + a_n^2}$$

2 ## The direction is not radial

When a tangential component exists the total acceleration is tilted away from the centre, by an angle whose tangent is a_t over a_n. Only at constant speed does it point exactly at the centre.

$$\tan\phi = \frac{a_t}{a_n}$$

→ braking on a curve means the acceleration points backward and inward

3 ## The reverse question

Given the total acceleration and one component, recover the other and then the radius. This is the standard numerical form: a car has 13 m/s squared total and 5 m/s squared tangential at 10 m/s, so the normal part is 12 and the radius is 8.3 m.

$$a_n = \sqrt{a^2 - a_t^2}, \qquad R = \frac{v^2}{a_n}$$

4 ## A worked curve

A car at 18 m/s brakes at 3 m/s² while on a 40 m radius curve. Find the total acceleration and its angle from the radius.

$$a_n = \frac{324}{40} = 8.1\ \text{m/s}^2, \qquad a = \sqrt{9 + 65.6} = 8.64\ \text{m/s}^2$$

$$\phi = \tan^{-1}\frac{3}{8.1} = 20.3^\circ$$

→ the car is slowing and turning at once, and the vector says which dominates

KEY RESULT

the two components

$$a_t = \dot{v}, \qquad a_n = v^2/R$$

NUMBERS

a check

v = 10, a_t = 5, a = 13

a_n = 12, R = 8.3 m

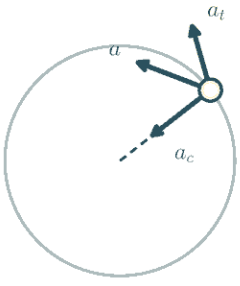
TRAP

using R from the total acceleration

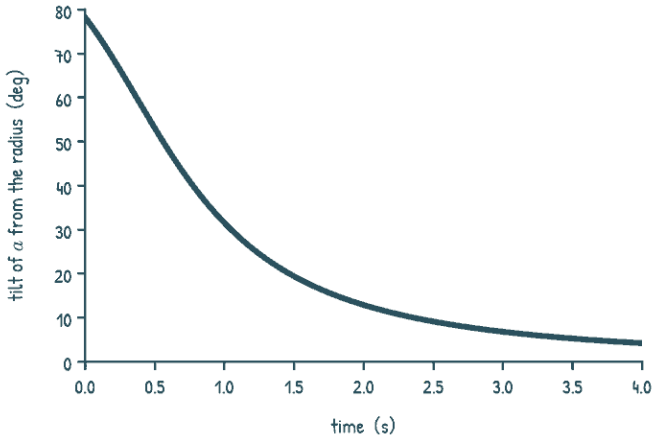
Only the normal component sets the radius. The tangential part does not bend the path at all.

at_an

$$a = \sqrt{a_t^2 + a_c^2}$$



as it speeds up, a turns radial



KEEP / two perpendicular components: one changes the speed, one changes the direction

What if the angular acceleration is not constant?

? A wheel slows under a torque that falls with time. The three rotation equations fail. What replaces them?

1 Back to the definitions

When alpha varies, go back to the derivatives. Integrating alpha over time gives the change in omega, and integrating omega gives the change in angle. Each integral is an area under a graph.

$$\omega = \omega_0 + \int_0^t \alpha\, dt, \qquad \theta = \theta_0 + \int_0^t \omega\, dt$$

2 The other chain

If alpha is given as a function of the angle instead of the time, use the rotational twin of v dv/dx. It removes the clock and often turns a hard integral into an easy one.

$$\omega\, d\omega = \alpha\, d\theta \quad \Rightarrow \quad \tfrac{1}{2}\omega^2 = \int \alpha\, d\theta$$

→ the same trick as a = v dv/dx, in rotation

3 A worked shape

For a wheel slowing under friction with alpha proportional to minus omega, the solution is exponential decay of the angular speed: each second removes the same fraction of what is left. That is the rotational analogue of linear drag.

$$\alpha = -k\omega \;\Rightarrow\; \omega = \omega_0 e^{-kt}$$

4 A worked decay

A wheel slows with alpha = −0.8 omega rad/s². From 25 rad/s, find omega at 2 s and the total angle before stopping.

$$\omega = 25e^{-1.6} = 5.05 \text{ rad/s}$$

$$\theta = \frac{\omega_0}{k} = \frac{25}{0.8} = 31.3 \text{ rad} = 5.0 \text{ turns}$$

→ an exponential spin-down never quite stops, but the angle converges

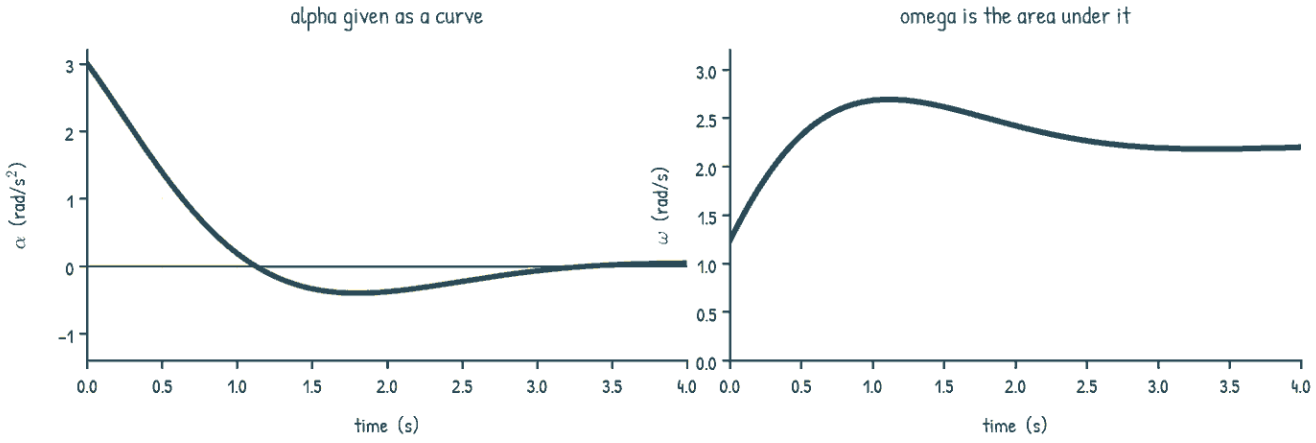
KEY RESULT
the integrals

$$\omega = \int \alpha\, dt, \qquad \theta = \int \omega\, dt$$

GATE
choose the variable
alpha of time: integrate over t. alpha of angle: use omega d omega = alpha d theta.

TRAP
reusing omega_0 + alpha t
That equation assumes constant alpha. Check before writing it, and if alpha is a function, integrate.

alpha_integral



KEEP / integrate: omega is the area under alpha, theta the area under omega

How fast do two points on a wheel separate?

? Two particles move on the same circle at different angular rates. How often do they meet, and what is their relative speed?

1 ## Subtract the rates

Both particles share the same circle, so their separation angle changes at the difference of the angular velocities. Meetings happen once per full relative revolution, which gives a period of 2 pi over the difference.

$$\omega_{\text{rel}} = \omega_1 - \omega_2, \qquad T_{\text{meet}} = \frac{2\pi}{|\omega_1 - \omega_2|}$$

2 ## Relative speed along the circle

The separation grows along the arc, so the arc grows at the relative angular rate times the radius. For two independent particles on one circle this is the closing speed along the track; for two points of one rigid wheel the rates are equal and the gap never changes.

$$v_{\text{rel}} = R\,|\omega_1 - \omega_2|$$

→ a rigid wheel spins its points together: a shared rate means a fixed gap

3 ## The same idea across frames

If the whole circle is also rotating, add or subtract the frame rate first. This is the rotational version of relative velocity, and it is what makes gear trains and pulleys tractable: the contact points share a velocity, not a rate.

4 ## A worked meeting

Two runners on a 200 m track go at 6 and 4.5 m/s the same way. The track length, not the radius, is given. When do they next meet?

$$R = \frac{200}{2\pi} = 31.8\ \text{m}, \quad \omega_1 = 0.188,\ \omega_2 = 0.141\ \text{rad/s}$$

$$T_{\text{meet}} = \frac{2\pi}{0.0471} = 133\ \text{s} = 2.2\ \text{min}$$

→ opposite ways the rates add: they then meet every 19 s

KEY RESULT

relative rotation

$$T_{\text{meet}} = 2\pi/|\omega_1 - \omega_2|$$

NUMBERS

a check

omega = 1.5 and 0.6 rad/s

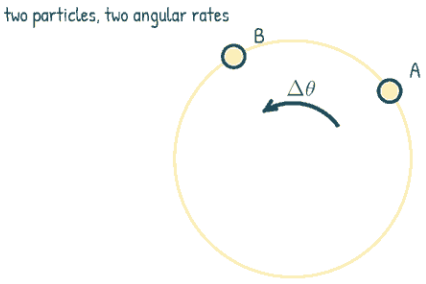
T_meet = 2 pi / 0.9 = 7.0 s

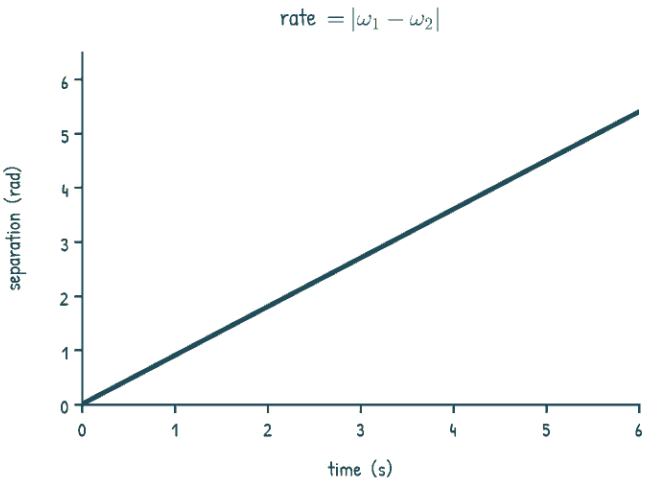
TRAP

adding the rates

Same-direction rotation subtracts. Adding gives the answer for opposite directions.

relative_angular





KEEP / relative angular rate is the difference, and relative speed is that times the radius

If the centripetal force is always inward, how does anything speed up?

? On a vertical circle gravity has a tangential component; the tension never does. Which force changes the speed, and which only bends the path?

1 ## Perpendicular means zero work

The centripetal force is along the radius and the displacement is along the tangent, so the dot product vanishes at every instant. It therefore cannot change the kinetic energy, which is why a car on a flat curve keeps a constant speed while its velocity turns.

$$W_c = \int \mathbf{F}_c \cdot d\mathbf{r} = 0$$

2 ## Who speeds the body up

Only the tangential component does work. On a vertical circle that is the tangential part of gravity, m g sin of the angle from the lowest point, so the work done by gravity is exactly minus the change in m g h. On a banked curve it is friction along the path.

$$m\tfrac{dv}{dt} = F_{\text{tangential}} = mg\sin\phi$$

3 ## Why the two equations are separate

The radial equation fixes the force needed for the instantaneous geometry; the tangential equation fixes how the speed evolves. Solving them in that order is the standard approach, and it is the reason energy can replace the tangential equation entirely.

→ energy handles the tangential direction; the radial equation handles the normal one

4 ## A worked split

A 0.5 kg bead on a 0.6 m string is at 4 m/s when the string makes 60° with the vertical, moving up. Find the tangential acceleration and the tension.

$$a_t = g\sin 60^\circ = 8.49\ \text{m/s}^2$$

$$T = \tfrac{0.5(16)}{0.6} + 0.5(9.8)\cos 60^\circ = 13.3 + 2.45 = 15.8\ \text{N}$$

→ gravity does the speeding up or slowing down; the string only bends the path

KEY RESULT

the split

radial: sets the path, no work. tangential: sets the speed, does the work.

TRAP

tension doing work

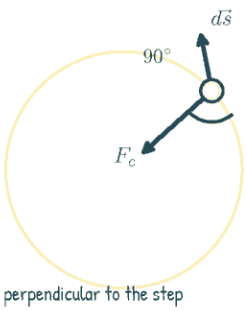
For a string of fixed length the tension is perpendicular to the motion and does no work. A string that changes length, or a peg that moves, can.

INSIGHT

the energy shortcut

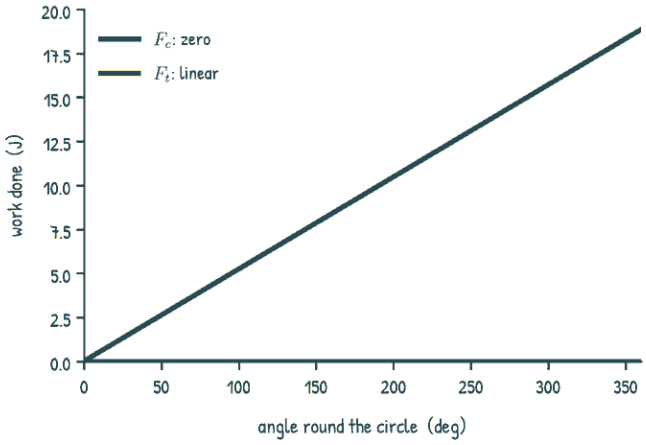
Because the radial force is powerless, the whole tangential problem collapses to conservation of mechanical energy.

work_radial



the radial force is always perpendicular to the step

only the tangential part earns work



KEEP / a radial force does no work, so only the tangential component can change the speed

What speed does a bead have at the top of a loop?

? A bead is released from rest at height h above the bottom of a smooth vertical loop. How fast is it at the top, and when does it leave the track?

1 Energy from the release point

With no friction the speed at any angle follows from the height dropped. Measuring the angle from the bottom, the height gained is R times one minus cosine, and the speed follows directly.

$$v^2 = v_0^2 - 2gR(1-\cos\phi)$$

2 The track condition

For a bead on the inside of a track the normal must stay positive. Setting it to zero gives the minimum speed at each angle, and the most demanding point is the top, where gravity already supplies the whole centripetal requirement.

$$N = \frac{mv^2}{R} + mg\cos\phi \;\geq 0$$

→ at the top cos phi = −1, so N = m v squared / R − m g

3 The two thresholds

For a full loop without leaving the track the top speed must satisfy v squared at least g R, which needs a release height of two and a half radii above the bottom. For a bead on a rod, or a bead threaded on a wire, there is no such condition and the minimum is only the height of the top.

$$h_{\min} = 2.5R \text{ (track)}, \qquad h_{\min} = 2R \text{ (rod)}$$

4 A worked loop

A 0.2 kg bead is released from rest 1.5 m above the bottom of a 0.5 m loop. Find the speed at the top and say whether it stays on.

$$v^2 = 2(9.8)(1.5-1.0) = 9.8 \;\Rightarrow\; v = 3.13 \text{ m/s}$$

$$v_{\text{top, min}}^2 = gR = 4.9 \;:\; 9.8 > 4.9, \text{ it stays on}$$

→ the margin is exactly a factor of two in v squared

KEY RESULT

the pair

$$v^2 = v_0^2 - 2gR(1-\cos\phi)$$

$$N = \frac{mv^2}{R} + mg\cos\phi$$

NUMBERS

a check

R = 0.8 m, released from rest at 2.5R

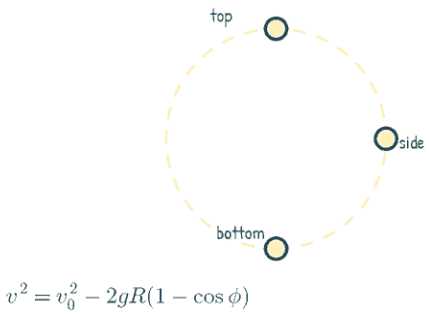
v_top = sqrt(gR) = 2.8 m/s, N = 0 there

TRAP

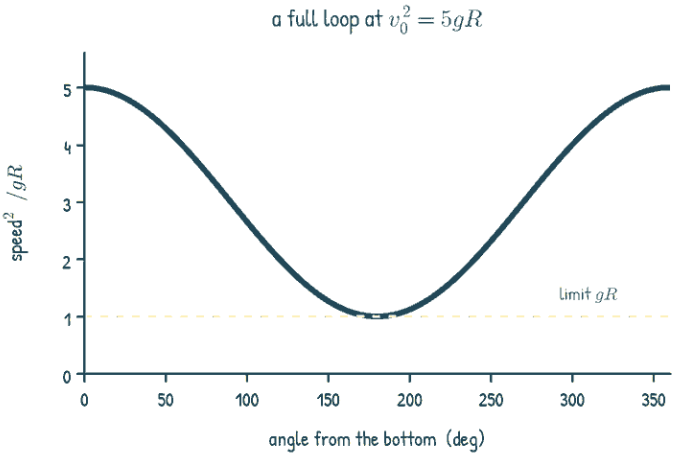
using v = 0 at the top

A bead on the inside of a track needs sqrt(gR) at the top, not zero. Zero is the rod answer.

vertical_energy



$v^2 = v_0^2 - 2gR(1-\cos\phi)$



KEEP / energy gives the speed at every angle; the track condition decides whether it stays on

Why must a string be launched faster than a rod?

? The same mass, radius and release height. On a string it must be launched at sqrt(5gR); on a rod, sqrt(4gR) is enough. Where does the difference come from?

1

The constraint is one-sided

A string can pull but not push, so the tension must remain non-negative everywhere. A rod can supply either sign, so the only requirement is that the bead reaches the top with enough speed to keep going.

$$T \geq 0 \text{ (string)}, \qquad N \text{ any sign (rod)}$$

2

Both boundary cases

At the top of a string loop the tension just vanishes at the limiting speed, giving v squared equal to g R there and 5 g R at the bottom by energy. For a rod the limiting speed at the top is zero, which gives 4 g R at the bottom.

$$v_{\text{bot}} = \sqrt{5gR} \text{ (string)}, \qquad \sqrt{4gR} \text{ (rod)}$$

→ the factor sqrt(5/4) = 1.12 in speed is a factor 1.25 in energy

3

What the tension does along the way

For the string at the limiting launch speed the tension is zero at the top and 6 m g at the bottom. For the rod the normal force changes sign somewhere above the horizontal, and the bead is being pushed inward there rather than pulled.

$$T_{\text{bot}} = 6mg \text{ (string, limiting)}, \qquad N_{\text{top}} = -mg \text{ (rod, limiting)}$$

4

A worked comparison

A 0.2 kg ball on a 0.9 m string is whirled in a vertical circle. Find the minimum bottom speed and the tension there.

$$v_{\text{bot}} = \sqrt{5(9.8)(0.9)} = 6.64 \text{ m/s}$$

$$T_{\text{bot}} = 6mg = 11.8 \text{ N}$$

→ a rod would need only 5.94 m/s, and a bottom force of 5mg

KEY RESULT

the two minima

$$v_{\text{bot}} \geq \sqrt{5gR} \text{ (string)}, \qquad \geq \sqrt{4gR} \text{ (rod)}$$

NUMBERS

a check

R = 0.8 m: string 6.26 m/s, rod 4.85 m/s

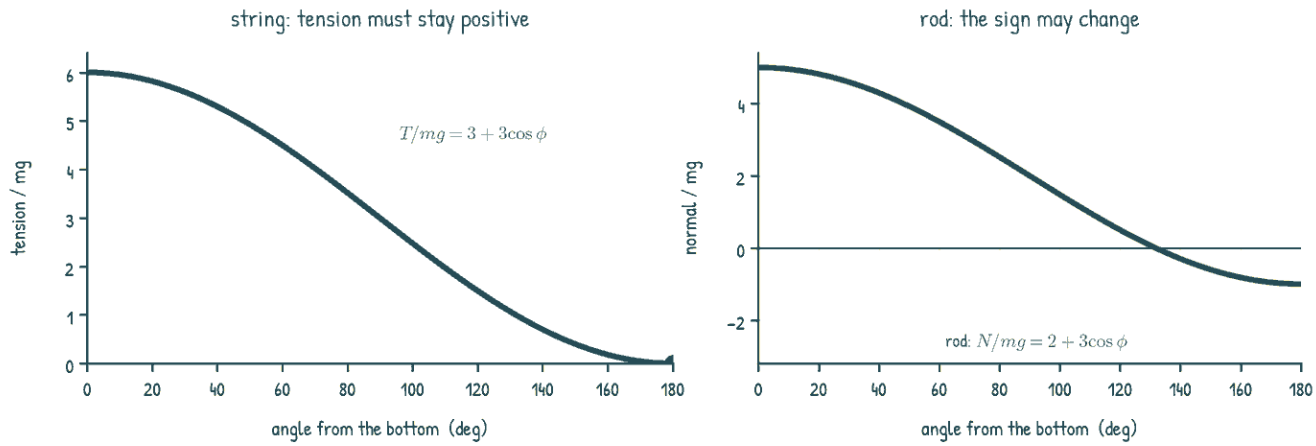
tensions at the bottom: 6mg and 5mg for the two limits

TRAP

the source 3mg

The bottom tension at the limiting speed is 6mg, not 3mg. See the source audit for the corrected figure.

string_vs_rod



KEEP / a string can only pull, so its tension must stay positive; a rod can push

Why does a conical pendulum ignore the mass?

? A bob whirls in a horizontal circle on a string of length L at angle theta. The tension depends on the mass, yet the period does not. Why?

1 ## Two equations at once

Vertically the vertical component of the tension balances the weight. Horizontally the horizontal component supplies the centripetal force. Divide one by the other and the mass and the tension both disappear.

$$T\cos\theta = mg, \qquad T\sin\theta = \frac{mv^2}{L\sin\theta}$$

2 ## What survives

The division leaves the angle fixed by the speed, and the period depends only on the vertical length of the string. Every bob on the same string length and the same angle has the same period, whatever its mass.

$$v^2 = gL\sin\theta\tan\theta, \qquad T = 2\pi\sqrt{\frac{L\cos\theta}{g}}$$

→ the vertical projection L cos theta is the length that matters

3 ## The limits

As the angle goes to zero the motion becomes a small planar pendulum with period 2 pi root L over g. As the angle approaches 90 degrees the required speed diverges and the period falls to zero: the string becomes horizontal only at infinite speed.

$$\theta \to 0:\ T \to 2\pi\sqrt{L/g}, \qquad \theta \to 90^\circ:\ v \to \infty$$

4 ## A worked whirl

A 1.2 m string carries a 0.3 kg bob at 30° to the vertical. Find the tension, speed and period.

$$T = \frac{0.3(9.8)}{\cos 30^\circ} = 3.40\ \text{N}$$

$$v = \sqrt{9.8(1.2)\sin 30^\circ \tan 30^\circ} = 1.84\ \text{m/s}$$

$$T_{\text{period}} = 2\pi\sqrt{1.2(0.866)/9.8} = 2.05\ \text{s}$$

→ the same string at 60° would need 2.4 times the speed and swing faster

KEY RESULT

the conical pendulum

$$T = 2\pi\sqrt{L\cos\theta/g}, \qquad v = \sqrt{gL\sin\theta\tan\theta}$$

NUMBERS

a check

L = 0.5 m, theta = 40°, m = 0.2 kg

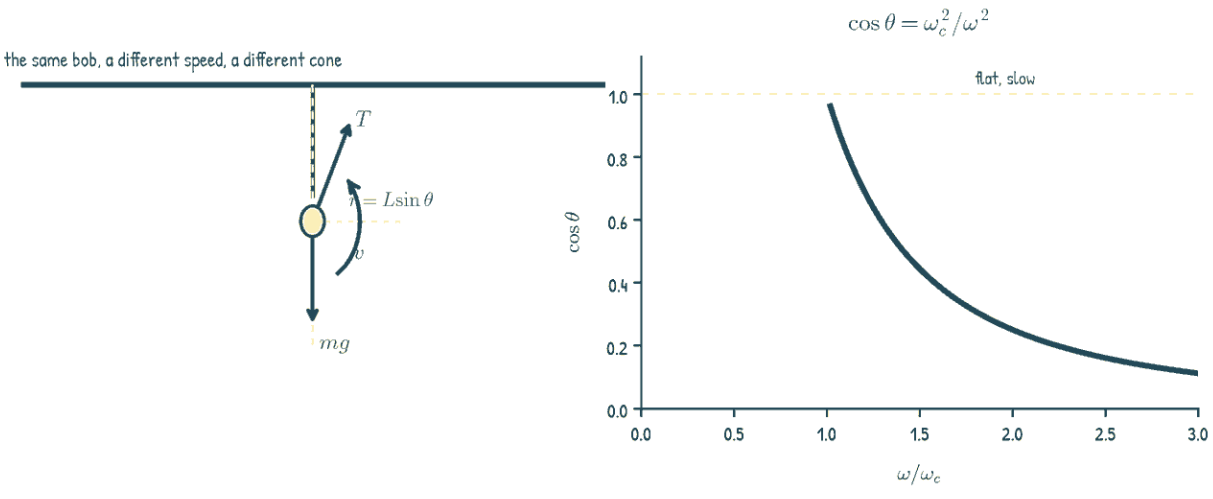
T = 1.24 s, v = 1.63 m/s, tension 2.56 N

TRAP

using L in the period

The period uses L cos theta, the vertical drop, not the string length.

conical_energy



KEEP / the two force equations both carry m, so it cancels out of the geometry and the period

What is the slowest safe speed on a banked corner?

? Banking is designed for one speed. Too fast and the car slides up; too slow and it slides down. What sets both limits?

1 ## Design speed

With no friction the normal force alone must supply the centripetal force, and resolving along and perpendicular to the bank gives the design condition.

$$\tan\theta = \frac{v_0^2}{Rg}$$

2 ## The two extremes

At the maximum speed friction points down the slope, helping the bank turn the car. At the minimum it points up the slope, stopping the car sliding down. Both come from the same resolution with the sign of friction flipped.

$$v_{\max}^2 = Rg\frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}$$

$$v_{\min}^2 = Rg\frac{\sin\theta - \mu\cos\theta}{\cos\theta + \mu\sin\theta}$$

3 ## Read the structure

Both reduce to tan theta when mu vanishes, the design speed. The maximum grows without bound as mu approaches cot theta, while the minimum falls to zero once mu exceeds tan theta.

$$\mu > \tan\theta \;\Rightarrow\; v_{\min} = 0$$

→ the band is asymmetric: friction helps much more at the fast end

4 ## A worked band

A 120 m radius corner is banked at 15° with mu_s = 0.45. Find both speed limits.

$$v_{\max}^2 = 1176\frac{0.259 + 0.435}{0.966 - 0.117} = 963 \;\Rightarrow\; 31.0\ \text{m/s}$$

$$v_{\min}^2 = 1176\frac{0.259 - 0.435}{0.966 + 0.117} < 0 \;\Rightarrow\; v_{\min} = 0$$

→ mu exceeds tan 15° = 0.268, so the car can stand still on the bank

KEY RESULT

the band

$$v_{\max}^2 = Rg\frac{\sin\theta + \mu\cos\theta}{\cos\theta - \mu\sin\theta}$$

and the min with the signs flipped

NUMBERS

a check

R = 80, theta = 27°, mu = 0.35

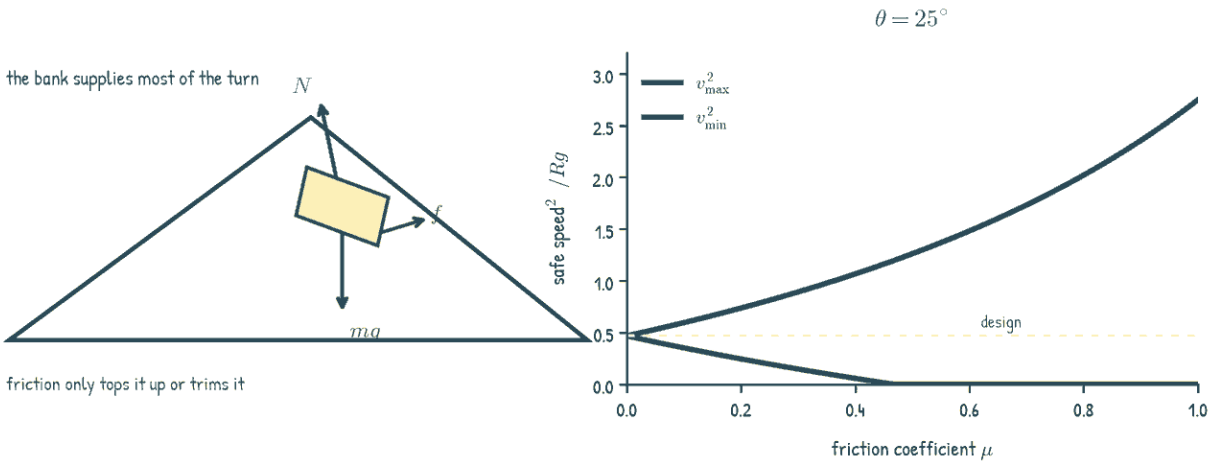
v: 10.3 to 28.6 m/s, design 20 m/s

TRAP

one formula for both

The friction direction flips at the design speed. Using one form for both limits gives a wrong minimum.

bank_limits



KEEP / friction can only widen the band around the design speed; it cannot create one

How does a wall of death hold a rider up?

? A rider circles the inside of a vertical cylinder. The wall is vertical and there is no floor under the tyres. What holds the weight?

1

Two perpendicular jobs

The normal force is horizontal, points at the axis, and provides the centripetal force. Friction is vertical and holds the weight. Neither can do the other job, so the two equations are independent.

$$N = \frac{mv^2}{R}, \qquad f = mg$$

2

The minimum speed

Static friction has a ceiling proportional to the normal. Substituting the two equations gives the minimum speed at which the wall can still hold the rider, and it falls as the friction coefficient grows.

$$\mu_s \frac{mv^2}{R} \geq mg \quad \Rightarrow \quad v_{\min} = \sqrt{\frac{gR}{\mu_s}}$$

→ a smaller cylinder, or a stickier wall, lowers the required speed

3

Why the rider feels heavy

At the minimum speed the friction is exactly at its limit, so any bump or lean that reduces the normal drops the rider. That is why the manoeuvre is done at well above the minimum, and why the normal force on the rider can be several times the weight.

4

A worked rotor

A 70 kg rider circles a 5 m radius wall of death at mu_s = 0.6. Find the minimum speed and the normal force then.

$$v_{\min} = \sqrt{9.8(5)/0.6} = 9.04 \text{ m/s}$$

$$N = \frac{70(81.7)}{5} = 1.14 \text{ kN}, \qquad f = 686 \text{ N}$$

→ at the minimum speed the friction is exactly at its ceiling

KEY RESULT

the rotor

$$v_{\min} = \sqrt{gR/\mu_s}$$

normal horizontal, friction vertical

NUMBERS

a check

R = 4 m, mu_s = 0.4

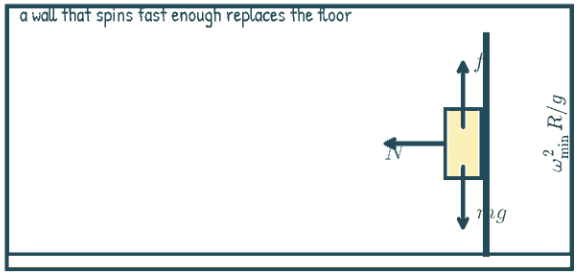
v_min = sqrt(98) = 9.9 m/s, about 36 km/h

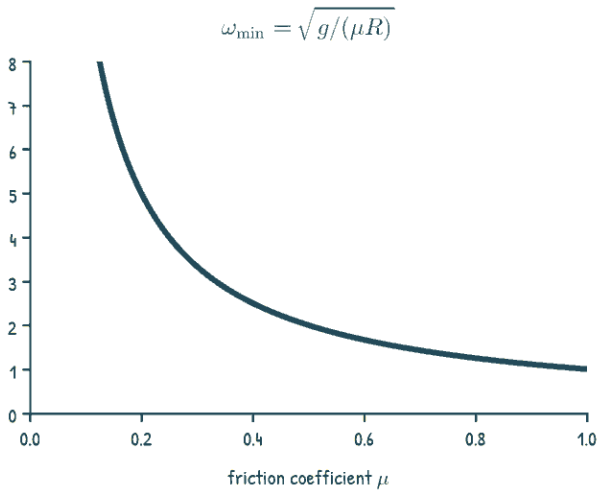
TRAP

friction supplying the centre

On a vertical wall friction cannot point at the axis. The normal does that; friction is vertical.

rotor_wall





KEEP / the normal provides the centripetal force, and static friction on the vertical wall carries the weight

How tight is a curve that is not a circle?

? A projectile path, a roller-coaster dip, a road bend: none is a circle. What is the radius at a point?

1 Define it by the normal acceleration

At any point of any smooth path there is one circle that matches it to second order. Its radius is the radius of curvature, and the normal component of acceleration is v squared over that radius.

$$\rho = \frac{v^2}{a_\perp}$$

2 Projectile values worth knowing

At the apex the velocity is horizontal and the whole of g is normal, so the radius is v squared over g with the horizontal speed. At launch the normal component is g cos theta, which gives a larger radius.

$$\rho_{\text{apex}} = \frac{v_0^2 \cos^2\theta}{g}, \qquad \rho_{\text{launch}} = \frac{v_0^2}{g\cos\theta}$$

→ for a vertical launch the path is straight and the radius is infinite

3 The general formula

For a path given as y of x, the radius follows from the first and second derivatives. A straight line has zero second derivative and infinite radius; a sharp bend has a large second derivative and a small radius.

$$\rho = \frac{(1 + y_x^2)^{3/2}}{|y_{xx}|}$$

→ at the apex of a parabola y double dash is g over v_x squared, which reproduces the result above

4 A worked curvature

A projectile is launched at 25 m/s at 40°. Find the radius of curvature at the apex and at launch.

$$\rho_{\text{apex}} = \frac{625\cos^2 40^\circ}{9.8} = 37.3 \text{ m}$$

$$\rho_{\text{launch}} = \frac{625}{9.8\cos 40^\circ} = 83.2 \text{ m}$$

→ the path bends hardest where it is least steep and slowest in the horizontal

KEY RESULT

curvature

$$\rho = v^2 / a_\perp$$

and rho = (1 + y_x^2)^(3/2) / |y_xx| for a graph

NUMBERS

a check

20 m/s at 45°: apex rho = 200/9.8 = 20.4 m

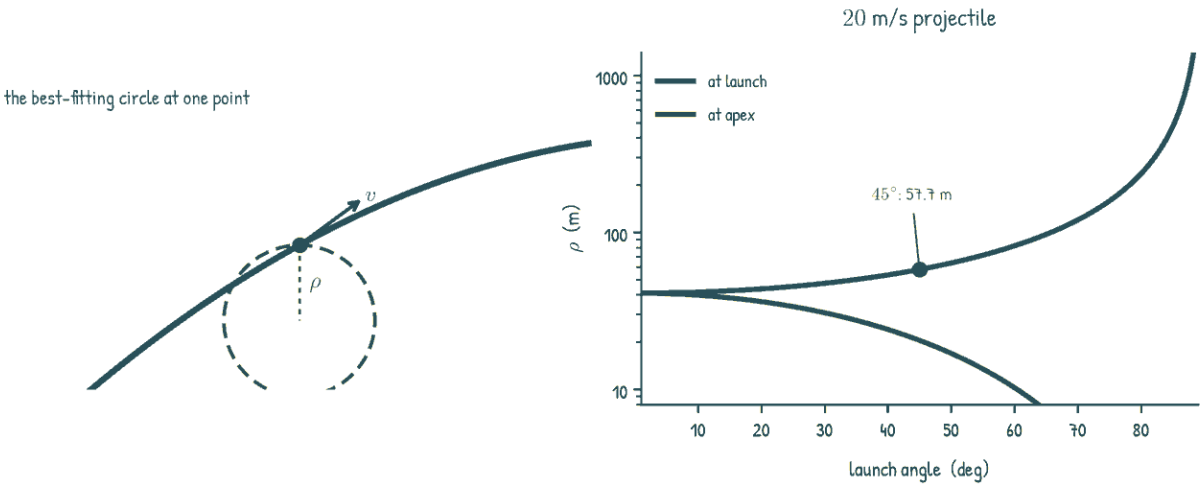
launch rho = 400/(9.8 x 0.707) = 57.7 m

TRAP

using the full speed at the apex

The apex speed is v0 cos theta, not v0. Using v0 overstates the radius by 1/cos² theta.

curvature_fit



KEEP / the radius of curvature is the circle that matches the path locally, and v squared over it is the normal acceleration